Practical Guidelines

Electricity and Electronics EBN 111

2025 - Semester 1

Lecturers: Prof. X. Ye, Mr. S. Twala, Mr. A. Rohde

Assistant Lecturers: L. Hurter, J. Nepembe, C. van Zyl, J. Ramango, I. Schilz

Guideline document overview

This guideline document contains the relevant resources required to conduct the practical assessments for EBN 111. An overview of the content in this practical guide is as follows:

1. **Circuit components**
2. **LTspice simulations**
3. **Lab environment**

Contents

1 Circuit Components

1.1 Summary

The practicals make use of a set of special components. If you need to replace any of the components, please take careful note of the specifications and part numbers of the components. If you use different parts in the practicals, you will get incorrect answers even if you build the circuits correctly.

Table 1: List of special components needed for the practicals

Component	**Specifications**
Potentiometer	$5k\Omega$, 0.5 W
Light Dependent Resistor (LDR)	$4k\Omega$
Light Emitting Diode (LED)	Red, 5mm, Forward Voltage: 1.7V
Transistor	BC337-40, 800mA, 45 V, 3-Pin.
Operational Amplifier	UA741CP, 8-Pin PDIP
Pushbutton tactile switch	50mA @ 24 V, surface mount.
9V battery connector	

These components can be bought from any electronics store, either in person or online.
Online stores:

- RSComponents
- Mantech
- Micorobotics

Physical stores and Online:

- Electronics123
- Communica
- ElectroComp

1.2 Battery

Batteries are used to supply voltage to a circuit. In the EBN practicals, 9 V batteries are used and are equivalent to a 9 V independent voltage source. It will be important to measure the exact voltage provided by the battery, as the voltage may vary slightly from 9 V and will decrease over time when used. In this practical, the battery will be represented using a 9 V independent voltage source in both the circuit schematics and in LTSpice. Figure 1 shows the schematic symbol for a battery, the LTSpice equivalent part, and an image of the battery used for the EBN practicals. LTSpice will be explored in more detail in Section 2. The 9 V batteries can be connected to a breadboard circuit easily by using a battery clip connector, shown in Figure 1 so that the positive and negative terminals of the battery can easily be connected to the breadboard.

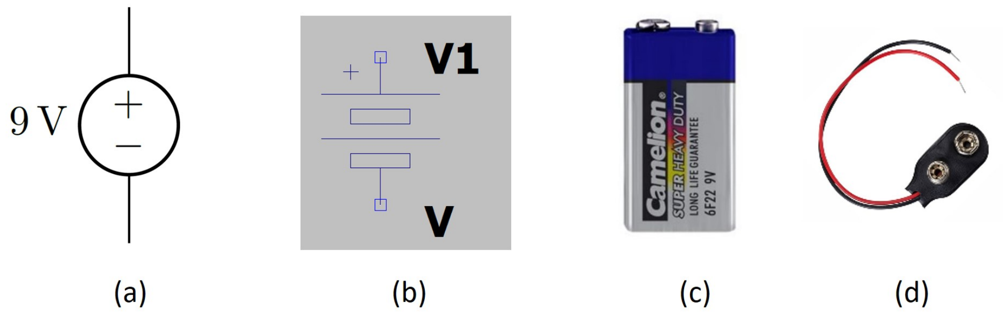

Figure 1: a) Schematic, b) LTSpice, c) image of battery component and d) image of battery clip connector used in the practical.

1.3 Resistor

A resistor is a passive electronic component (does not produce energy) that is used to implement an electrical resistance in a circuit. Resistors come in different shapes, sizes and materials. The resistors used in the practicals are carbon film resistors. Resistors do not have a polarity, so it does not matter which way you insert the resistor into your circuit (no positive and negative terminal). Figure 2 shows the symbol for a resistor, the LTSpice version and examples of some resistors with commonly used values.

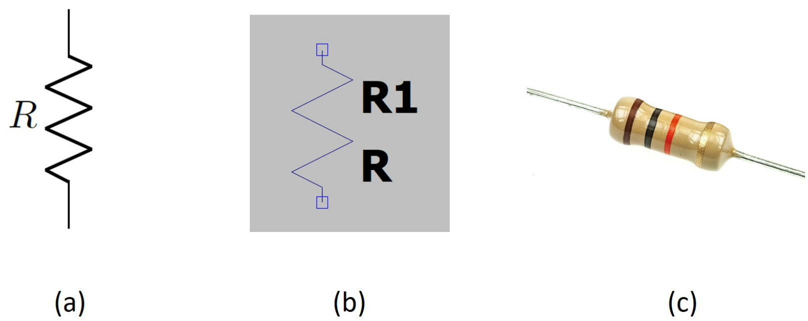

Figure 2: a) Schematic, b) LTSpice and c) Image of resistor components used in the practical.

Each carbon film resistor (such as those in your component boxes) has a specific resistance that has been set by the manufacturer. The value of this resistance is indicated by four coloured stripes on the resistor.

The first three stripes specify the resistance of the resistor (Figure 3). The first two stripes form the first two digits of the resistance value, this 2-digit value is then multiplied by the value of the 3rd stripe. The result is the resistance of the resistor.

The fourth stripe specifies the tolerance (Figure 4). A tolerance of 5 % (gold stripe) means that the actual resistance of the resistor can be any value within 5% of the value specified by the first three

Colour	Colour name	1st stripe (1st digit)	2nd stripe (2nd digit)	3rd stripe (multiplier)
	Black	0	0	x1
	Brown	1	1	x10
	Red	2	2	x100
	Orange	3	3	x1000
	Yellow	4	4	x10000
	Green	5	5	x100000
	Blue	6	6	x1000000
	Violet	7	7	-
	Grey	8	8	-
	White	9	9	-
	Gold	-	-	x0.1
	Silver	-	-	x0.01

Figure 3: Resistor colour codes for the first three stripes.

stripes. This deviation occurs because the manufacturing process is not perfect.

Colour	Colour name	4th stripe (Tolerance)
	Gold	5%
	Silver	10%

Figure 4: Resistor colour codes for the fourth stripe.

Figure 5 shows some examples of different resistor values. Note that a decimal point is often replaced by the SI prefix letter (e.g. 2.7 kΩ can be written as 2k7 Ω).

Figure 5: Examples of resistor colour codes from left to right: 1 kΩ (5%), 12 Ω (5%), and 5.6 kΩ or 5k6 Ω (5%).

Resistors are used to limit current in electronic circuits. An example of this is provided in Section 1.6 where a resistor can be used to limit the current flowing through an LED.

1.4 Potentiometer

A potentiometer is an electronic component with three pins that allows you to adjust the resistance of an electrical circuit. The three pins of a 3362 adjustable potentiometer are labelled 1, 2, and 3. To connect the three pins of a potentiometer, you need to understand how it works. A potentiometer is essentially a variable resistor that can be adjusted by turning a knob. The resistance between the left (pin 1) and centre (pin 2) pins of the potentiometer changes as you adjust the knob, which in turn affects the output of the circuit. Connect only pin 1 and pin 2 to your circuit, but do not bend or break pin 3. Once the potentiometer is connected to the circuit, you can adjust the resistance by turning the knob. Figure 6 shows the practical component of the 3362 potentiometer and its symbol.

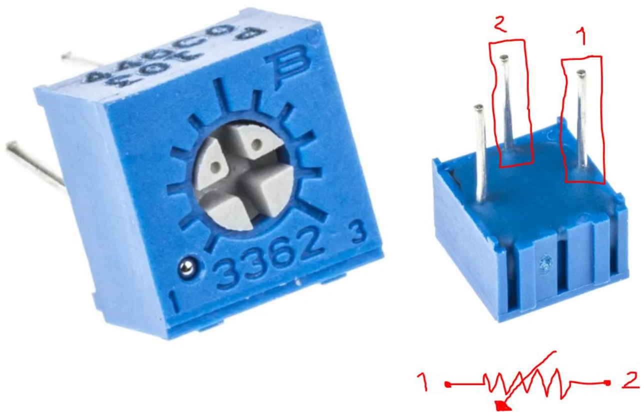

Figure 6: Image and schematic of a 3362 potentiometer component.

1.5 Light Dependent Resistor (LDR)

Resistors are used with transducers to make sensor subsystems. Transducers are electronic components, which convert energy from one form into another, where one of the forms of energy is electrical. A light dependent resistor, or LDR, is an example of an input transducer. Changes in the brightness of the light shining onto the surface of the LDR result in changes in its resistance. An input transducer is most often connected along with a resistor to make a voltage divider. In this case, the output of the potential divider will be a voltage signal, which reflects changes in illumination. Figure 7 shows the symbol for an LDR and an example of a practical LDR component.

Figure 7: a) Schematic, b) image of a typical LDR component.

1.6 Light Emitting Diode (LED)

An LED is a type of diode, which is a device that conducts current primarily in one direction. An LED is a light source that emits light when current flows through it. The current flows from the anode to the cathode of the LED to produce light, as shown in Figure 8. The anode is connected to the positive voltage supply (for example +9 V) and the cathode is connected to the negative voltage supply or ground (GND). An LED has a fairly narrow spectrum, and the human eye interprets the emitted light as a specific colour, depending on the peak wavelength of the LED. For this practical, a red LED is used (Figure 8).

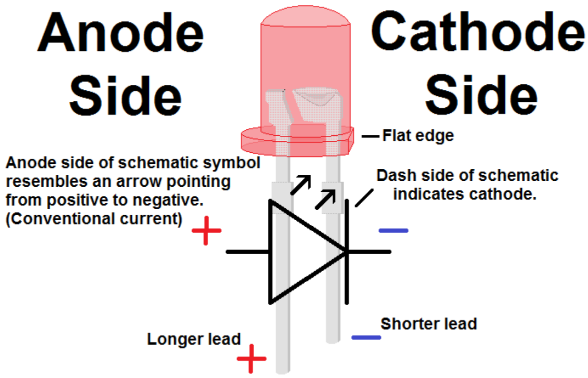

Figure 8: Schematic and image of an LED component.

The light emitted by an LED is directly proportional to the current through the LED. When an LED starts to conduct, the current increases rapidly and care must be taken not to apply too much current, as this can overheat and destroy the LED. The maximum current for the LED used in this practical is 20 mA. The current through the LED should not exceed this limit, with a general rule of using 50 % or less of the current limit, which still produces an acceptable brilliance of light. The LED will conduct when a specific voltage is applied across the LED, known as the forward voltage. The forward voltage is directly related to the wavelength, and thus differs depending on the colour of the LED. For the LED used in this practical, the forward voltage range is approximately 1.8 V.
By considering the current limit and forward voltage of the LED, a current limiting resistor can be used in series with the LED to ensure that the current through the LED is limited at all times to protect the LED from being damaged. A typical red LED requires a current of 10 mA and has a voltage of 1.8 V across it. The power supply to the circuit from the battery or the power supply is 9 V. Using Kirchoff's Voltage law, we can then determine the voltage which would sit across the resistor as: 9 V − 1.8 V = 7.2 V. For the current flowing through to be equivalent to 10 mA or less, using Ohm's law (V = RI), we can calculate the minimum value of the resistor to be: 7.2 V / 10 mA = 720 Ω. We select a resistor value of R1 = 510 Ω to ensure the current is well below the limit by still allowing the LED to light up when powered. Figure 9 shows the symbol for an LED in LTspice.

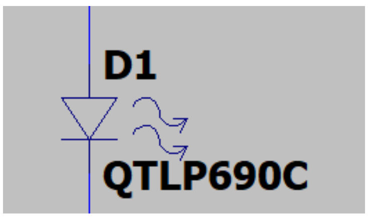

Figure 9: LTspice symbol of LED

1.7 Transistor

Transistors are essential components in modern electronics and are used in a wide range of applications, including amplification, switching, and digital logic. A transistor is a semiconductor device that can

amplify or switch electronic signals. In this section, we will provide a brief overview of transistors, including their operation, equivalent dependent source circuit, linear region versus the active region, and using the dependent source equivalent circuit versus using a practical transistor as a switch.
A transistor consists of three layers of semiconductor material, namely the emitter, base, and collector as seen in Figure 10.There are two types of bipolar junction transistors (BJTs) and field-effect transistors (FETs). In a PNP BJT, the current flow is controlled by the voltage applied to the base, while in an N-channel FET, the current flow is controlled by the voltage applied to the gate.

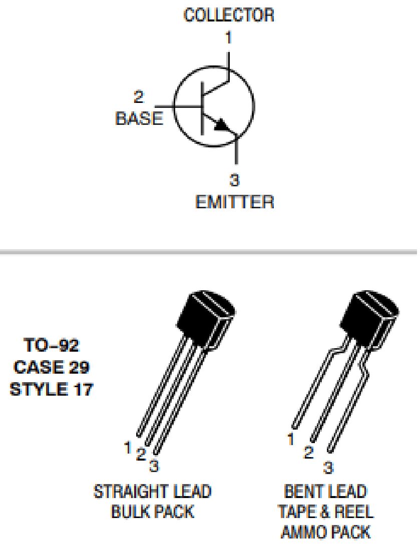

Figure 10: Schematic and image of a BC337-40 transistor

One way to model a transistor is to use an equivalent dependent source circuit. In this circuit, the transistor is modeled as a current source whose magnitude is proportional to the voltage applied to the base. The proportionality constant is known as the transistor's current gain, denoted by hFE. The current source is connected in series with a resistor, which models the transistor's internal resistance. The equivalent dependent source circuit is a useful model for analyzing transistor circuits, as it simplifies the circuit analysis process.
Transistors operate in two regions, namely the linear region and the active region. In the linear region, the transistor acts as an amplifier, where small changes in the input voltage result in proportional changes in the output current. In the active region, the transistor acts as a switch, where the output current is either fully on or fully off, depending on the input voltage.
When using the dependent source equivalent circuit, the transistor is typically operated in the linear region, where the output current is proportional to the input voltage. This is useful for amplifier circuits, where the goal is to amplify a weak signal. However, when using a practical transistor as a switch, it is typically operated in the active region, where the output current is either fully on or fully off, depending on the input voltage.
The main difference between using the dependent source equivalent circuit and using a practical transistor as a switch is that the dependent source equivalent circuit assumes that the transistor's current gain is constant, regardless of the operating conditions. However, in practice, the current gain varies depending on the operating conditions, such as temperature, voltage, and current. Therefore, when using a practical transistor as a switch, it is essential to consider the transistor's current gain and operating conditions to ensure reliable and consistent switching performance.

1.8 Switch

A switch is an important electronic component that can be used to turn a circuit on or off, or to control the flow of current in a circuit. A switch is a device that can be used to open or close an electrical circuit. When a switch is closed, it allows current to flow through the circuit. When it is open, it stops the flow of current.Switches come in many different types and configurations, but the most common types are:

1. Single-Pole, Single-Throw (SPST) switch (Figure 11): A switch that has only one set of contacts, which can be either open or closed.
2. Single-Pole, Double-Throw (SPDT) switch: A switch that has one input and two outputs. It can be used to connect the input to either output, but not both at the same time.
3. Double-Pole, Double-Throw (DPDT) switch: A switch that has two inputs and two outputs. It can be used to connect each input to either output, but not both at the same time.

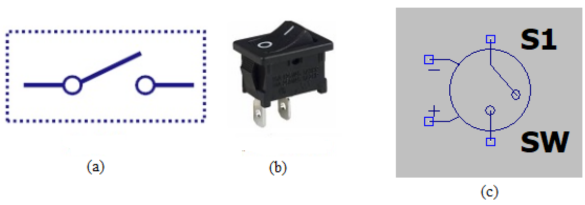

Figure 11: a) Schematic, b) image of a SPST switch, and c) LTspice symbol

1.9 Capacitor

Capacitors are passive electronic components that store and release electrical energy in a circuit. They consist of two conductive plates separated by a non-conductive material called a dielectric. When a voltage is applied to the capacitor, an electric field is generated between the two plates, causing charge to accumulate on each plate. The amount of charge that a capacitor can store depends on its capacitance, which is measured in farads (F).
Capacitors have many applications in electronic circuits, such as filtering out unwanted noise or voltage spikes, smoothing out signals, and tuning resonant circuits. They also play a vital role in energy storage and conversion in various electronic systems, such as power supplies, audio amplifiers, and electronic filters.
Capacitors come in a wide range of types and sizes, each with its own characteristics and properties. Some of the most common types of capacitors include ceramic capacitors, electrolytic capacitors, film capacitors, and tantalum capacitors. The choice of capacitor type and size depends on the specific application and requirements of the circuit. Figure 12 shows a single layer ceramic capacitor.

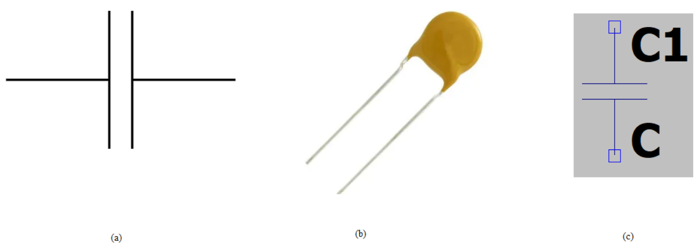

Figure 12: a) Schematic, b) image of a capacitor, and c) LTspice symbol

1.10 Operational Amplifier

Operational amplifiers, or op amps, are integrated circuits commonly used in analog electronic circuits. They are versatile components that can be used in a wide range of applications, from signal amplification to filtering to analog computation. In this section, we will introduce op amps and discuss their equivalent dependent source circuit, as well as the difference between using this circuit and using a practical op amp.

An op amp is a differential amplifier with a very high gain and a very high input impedance. It has two input terminals, labeled as the non-inverting (+) and inverting (-) inputs, and one output terminal. The input voltage difference between these two terminals, known as the differential input voltage, is amplified by the op amp to produce the output voltage. Figure 13 shows the schematic and the pinout connection of a typical 8-DIP op-amp.

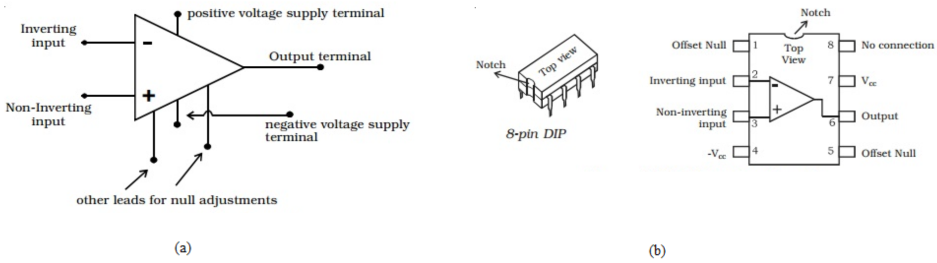

Figure 13: a) Schematic, b) image of a op-amp, and its pin configuration

The equivalent dependent source circuit of an op amp is a circuit model that simplifies the analysis of op amp circuits. It represents the op amp as a voltage-controlled voltage source (VCVS), with the gain of the op amp represented by the dependent voltage source's voltage gain. The input impedance of the op amp is represented by an infinite impedance connected between the two input terminals, and the output impedance is represented by a zero impedance connected to the output terminal.
It's essential to understand the linear region versus the active region of an op amp. The linear region of an op amp refers to the range of differential input voltages in which the output voltage is proportional to the input voltage. When the differential input voltage exceeds a certain value, the op amp enters the active region, where the output voltage saturates at the maximum or minimum voltage that the op amp can produce.
The dependent source equivalent circuit is useful for analyzing op amp circuits because it simplifies the math involved in analyzing the circuit's behavior. However, in practical circuits, op amps have limitations that are not represented in the dependent source equivalent circuit. For example, op amps have a limited input voltage range, a limited output voltage range, and a limited bandwidth. Additionally, op amps have non-ideal characteristics, such as input offset voltage, input bias current, and output impedance. These non-ideal characteristics can affect the performance of op amp circuits and must be considered in practical circuit design.

1.11 Breadboards or protoboards

Protoboards (or breadboards) are used to build prototype circuits. Component leads and wires are pushed into the holes in the protoboard to make electrical connections. See Section 3.1.1 for more details.

1.12 Probes and connectors

There are various probes and connector types that form part of the EBN component kits, with some examples provided in Section 1.12.1.

1.12.1 Crocodile clips, banana plugs and BNC plugs

1. The crocodile clips in Figure 14 (a) can be attached to a component lead, wire, or another crocodile clip for making electrical connections between the laboratory instruments and your circuit.

2. The banana plugs in Figure 14 (b) connect to the power supply and multimeter.

3. The BNC plugs in Figure 14 (c) connect to the function generator and oscilloscope.

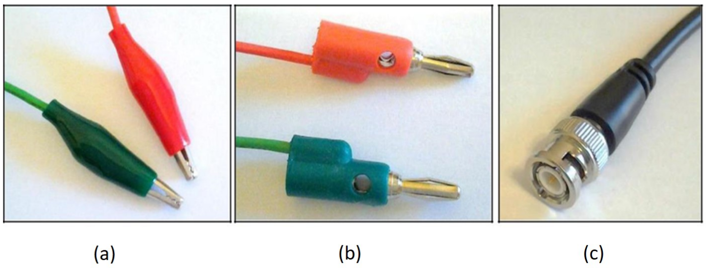

Figure 14: Examples of different connector types: (a) crocodile clips, (b) banana plugs and a (c) BNC plug.

1.12.2 High impedance probe

The probe in Figure 15 is used for monitoring high-frequency voltage signals on the oscilloscope. The crocodile clip part is generally connected to ground or 0V, and the probe part is placed at a node in the circuit where the desired voltage is measured. In other words, the probe will measure the potential difference between the probe and the crocodile clip.

Figure 15: High impedance, passive oscilloscope probe.

2 LTspice Simulation

2.1 Installing LTspice

1. Download the Windows or Mac version of the LTspice installation file from following link: https://www.analog.com/en/design-center/design-tools-and-calculators/ltspice-simulator.html The file is approximately 50 MB.

2. Click on the downloaded file *LTspice64.msi* to install LTspice.

3. Accept the licence agreement.

4. Click on *Install Now* and wait for the installation to complete.

5. When the installation is done, LTspice should launch automatically. If it does not, you can launch it by clicking on the *LTspice* shortcut in the start menu.

2.2 Creating a new schematic

1. When you open LTspice, launch it by clicking on the *LTspice* shortcut in the start menu. You should see the toolbar in Figure 16 at the top of the LTspice window. This toolbar and the drop-down menus contain all the tools you require for using LTspice.

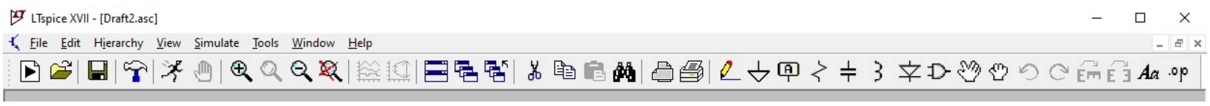

Figure 16: LTspice toolbar.

2. Before you can start building a new circuit, first create a new schematic in one of the following ways:

   (a) Click on *File* → *New schematic.*

   (b) Press *Ctrl + N.*

   (c) Click on the *New Schematic* icon shown below in Figure 17:

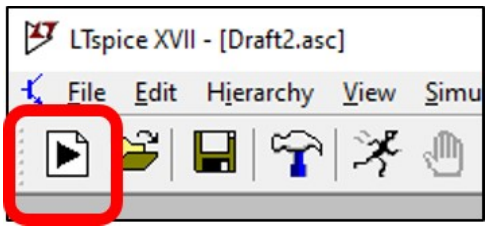

Figure 17: LTspice new schematic button.

   You will be presented with a blank grey schematic that you can use to build your circuit on.

2.3 Building a simple circuit

You will construct and simulate a simple circuit similar to the one required in the pre-practical assignments. To do so, follow the steps below:

1. Click on the *Component* button to add a new component as shown in Figure 18.

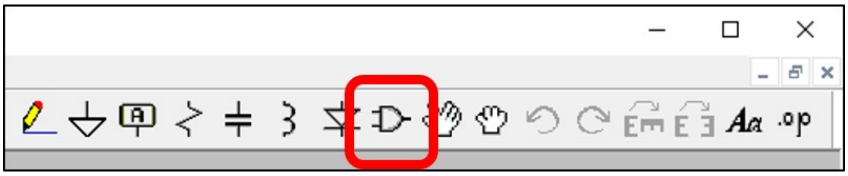

Figure 18: LTspice component button.

The *Select Component Symbol* window shown in Figure 19 will open. You can use this window to select and place components onto the schematic.

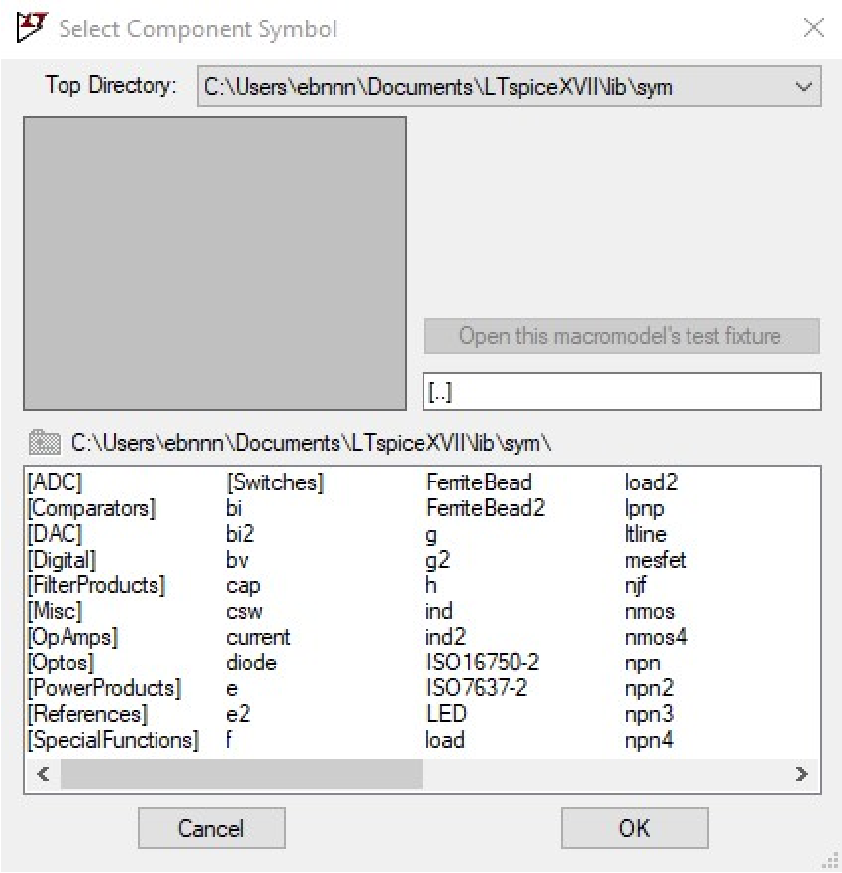

Figure 19: LTspice Select Component Symbol window.

The components listed are available for use in your circuits. The elements in square brackets are folders containing more components inside.
Note: The component names are often cryptic so you will have to explore and familiarize yourself with the component names.

2. Double click on *[Misc]* to open the miscellaneous folder. You will see the components listed in Figure 20.

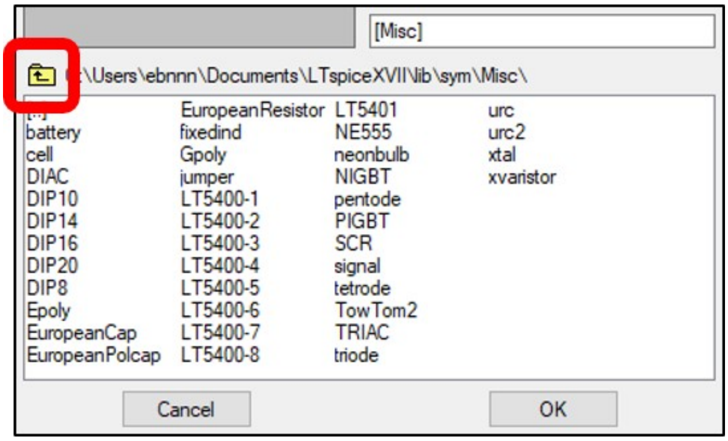

Figure 20: LTspice contents of miscellaneous folder.

Tip: To navigate back, click the *Up One Level* icon (folder icon) indicated in Figure 20 or click the *[..]* folder. Try to navigate the directories and make sure you are comfortable with the interface.

3. To add a DC voltage source, select the *battery* component in the *[Misc]* folder and click *OK*. Click anywhere on the schematic to place the component. You can place multiples of the selected component by clicking more than once. When you place one component, right-click to stop component placement, or press ESC on your keyboard. Your schematic will contain a battery and should look like Figure 21.

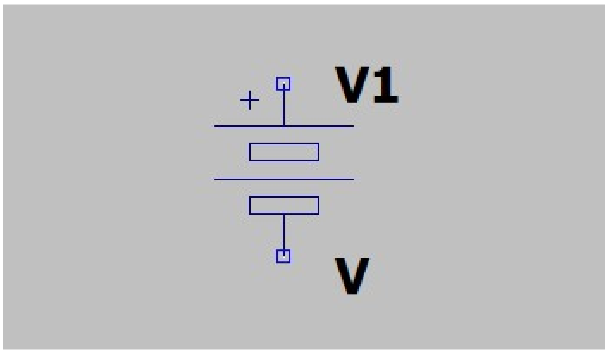

Figure 21: LTspice DC source (battery).

Tip: You can zoom in or out of the circuit by pressing *Ctrl* and rolling the mouse wheel.
Tip: You can pan the schematic by clicking and dragging anywhere on the schematic.
Note: Usually you would use the sources named *voltage* and *current* (find them in the home directory shown in Figure 19) since they can be set up to be a range of source types including DC, AC, pulse, and exponential. The sources are shown in Figure 22. The battery (Figure 21) is exactly the same as the voltage component (Figure 22 left) mentioned with a different symbol. You can choose which one to use in your circuits. The battery will be used for the example.

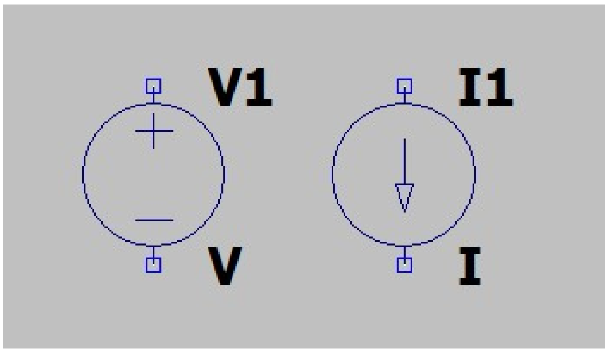

Figure 22: LTspice voltage and current sources.

4. Right-click on *V1*, the label of the DC source in Figure 21. You will see the window shown in Figure 23 which can be used to rename a component. Change the name of the DC source to *VS* and click *OK* (remember this name as it will be used later).

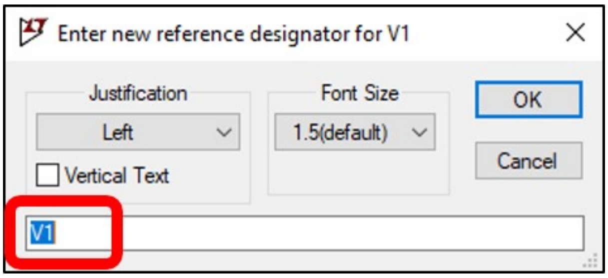

Figure 23: LTspice window used to rename a component.

Note: Two components may not have the same name or else the simulator will not know which is which and will result in an error.
Note: It is not required to rename components but it is very useful when building larger circuits.

5. Right-click on *V* (at the bottom of the component), the value of the DC source in Figure 21. A window similar to Figure 23 will open which can be used to set the value of a component. Change the value of the DC source to 12 V by entering 12 without any unit and click *OK*. The DC source should look like the one in Figure 24.

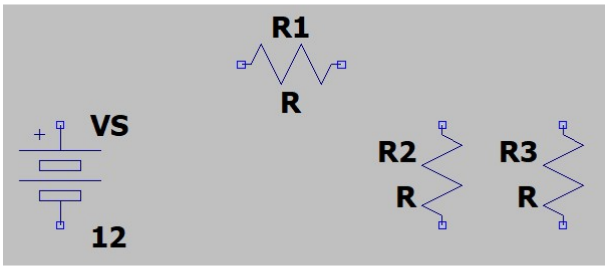

Figure 24: LTspice example of DC source with three resistors.

Note: When you right click on the component, there is a button labeled *Advanced*. Clicking this button will grant you access to a range of component parameters where you can set it up to be an AC, DC, or other source type.
Note: You may use SI prefixes such a “100m” for 0.1 V or “5k” for 5000 V when entering a value. Do not enter the unit (V) or any other combination of letters except for an SI prefix.

6. Place three resistors onto the schematic as shown in Figure 24. You can do this by clicking the *Component* button (Figure 18) and selecting the *res* component (you might have to go up one directory (see Figure 19)).
   Tip: You can rotate a component by pressing *Ctrl + R* while placing a component.
   Tip: You can move a component by pressing *F7* and clicking the component you want to move.
   Tip: You can delete a component by pressing *Delete* and clicking on the component.
   Tip: You can use keyboard shortcuts to make using LTspice easier. For example, to place a resistor you can press *R* on the keyboard. You can see more shortcuts by clicking on the *Edit* drop-down list as shown in Figure 25.

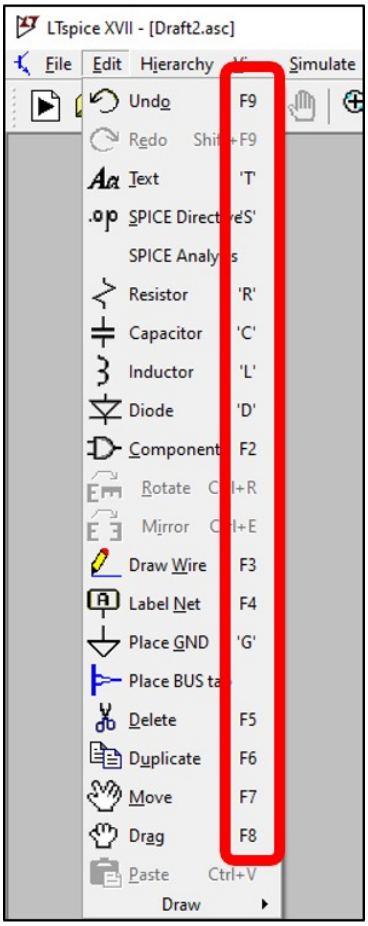

Figure 25: LTspice keyboard shortcuts.

7. Change the values of the three resistors to 10 Ω. You can do this by right-clicking the value (R) of each resistor in Figure 24, entering 10 in the popup window, and clicking *OK*. Your circuit should look like Figure 26.

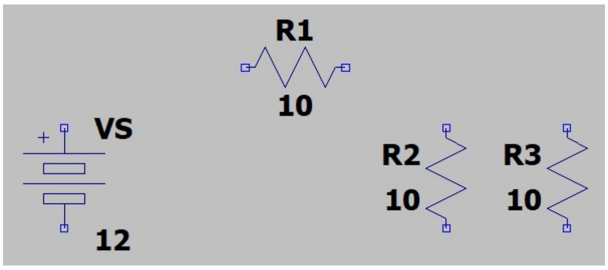

Figure 26: LTspice example with DC source and resistors with values defined.

8. To connect components together and complete the circuit, you need to place wires. You can place a wire by pressing *F3* or by clicking the *Wire* button shown in Figure 27.

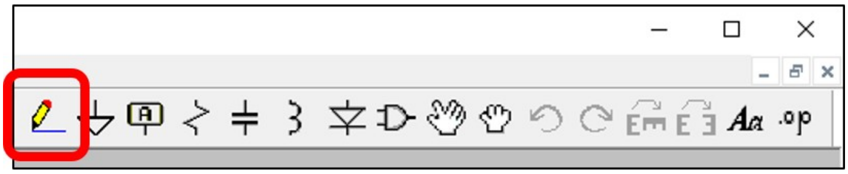

Figure 27: LTspice wire button.

Place the first wire by clicking the top node of the DC source indicated by the red cross in Figure 28. Drag the wire straight upwards and click to make a right turn. Drag the wire horizontally and connect it to the left node of the top-most resistor *(R1)*. Your circuit should look like Figure 28.

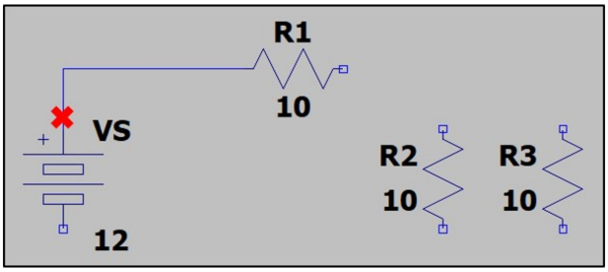

Figure 28: LTspice example of placing the first wire.

Tip: You can cancel wire placement by right-clicking or pressing Esc. You can delete incorrectly placed wires by pressing *Delete*.

9. Connect all of the components in your circuit as shown in Figure 29.

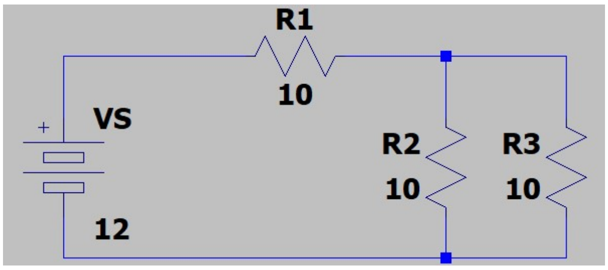

Figure 29: LTspice example of circuit with all components connected.

10. Before the circuit can be simulated, a ground reference is required. You can place a ground by pressing *G* or by clicking the button shown in Figure 30.

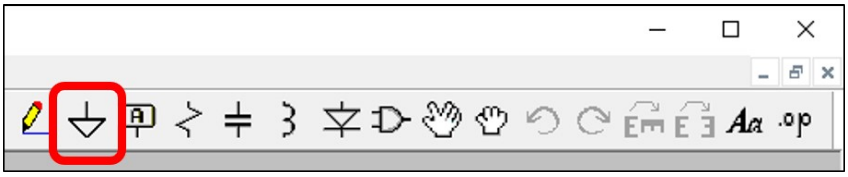

Figure 30: LTspice ground button.

Place the ground below the circuit and connect it to the bottom node of the circuit. Your circuit should look like Figure 31.

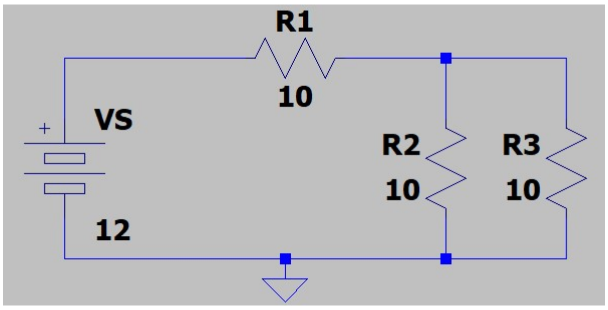

Figure 31: LTspice example of circuit with ground connection.

2.4 Running a Simulation

2.4.1 DC Operating Point Analysis

1. The circuit is now ready to simulate. The first step is to create a simulation profile. Click on *Simulate* at the top of the window to open a drop-down list as shown in Figure 32. Click on the *Edit Simulation Cmd* option. The *Edit Simulation Command* window shown in Figure 33 will appear.

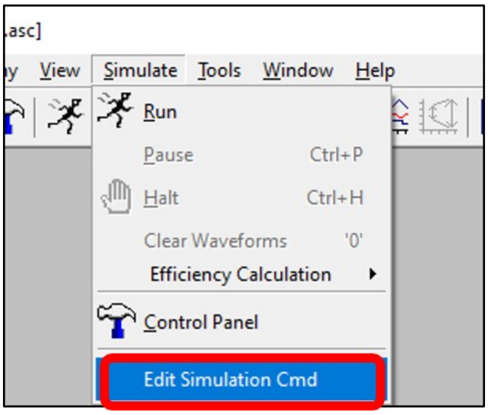

Figure 32: LTspice simulation drop-down list.

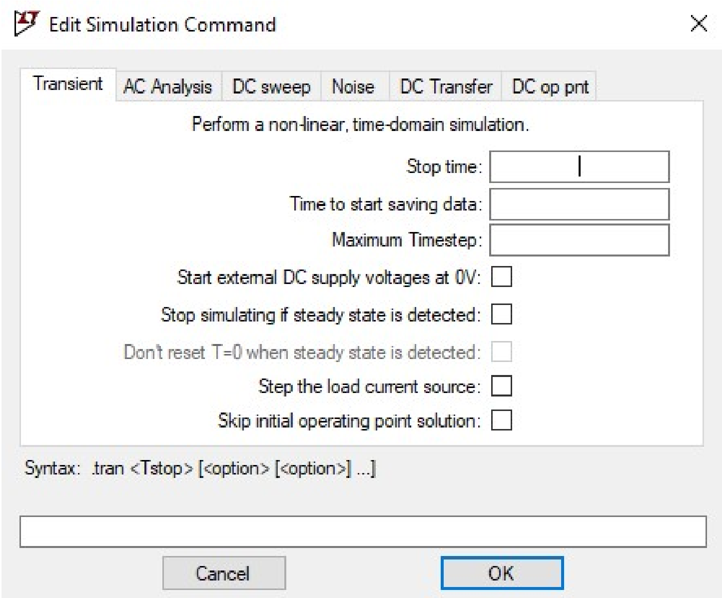

Figure 33: LTspice Edit Simulation Command window.

2. Select the kind of simulation you want to run. As a start, we are only interested in the steady-state DC node voltages and loop currents. Therefore, a DC operating point simulation will suffice. To set up the simulation, navigate to the DC operating point simulation type by clicking on the *DC op pnt* tab in the *Edit Simulation Command* window indicated in red in Figure 34. **Note:** You can choose different simulation types by navigating through the tabs. You will see the spice simulation command (blue box in Figure 34) change as you select different simulation types and change parameters. You can experiment with different simulation types on your own by using the user interface or by modifying the simulation command manually. You might need to do some Googling to understand some of the parameters.

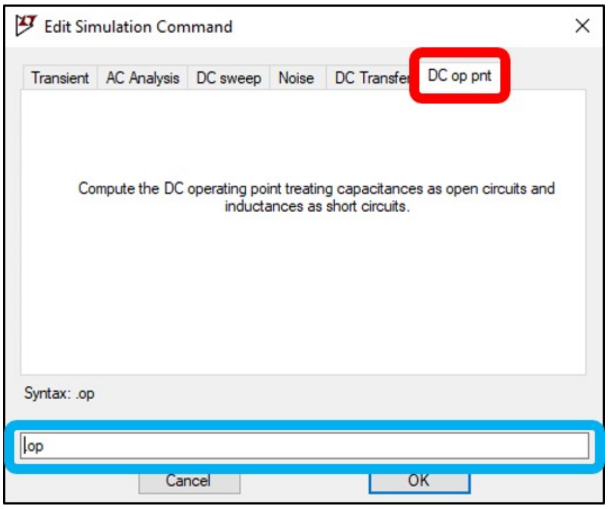

Figure 34: LTspice DC operating point tab.

3. Click *OK* to close the *Edit Simulation Command* window. You will see that some text now follows your cursor. Click somewhere on your schematic to place the text. This text is the spice simulation command that was also displayed in the blue box of Figure 34. This text will provide the software with all the parameters required to run the simulation. Your schematic should look like Figure 35 (note the simulation command in the bottom left).

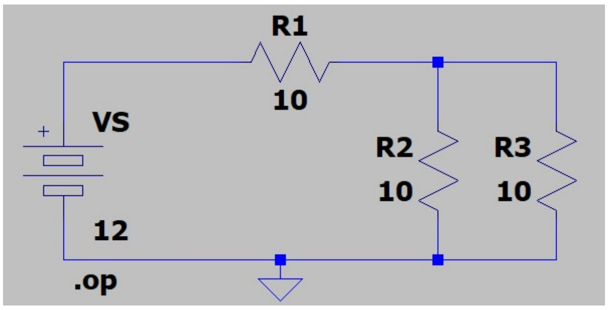

Figure 35: LTspice example of schematic with simulation command.

Tip: It is useful to understand how to interpret the parameters of the simulation command. This will make debugging your circuit and simulation much easier in the future.

4. To run the simulation, click on the *Run* button indicated in Figure 36.

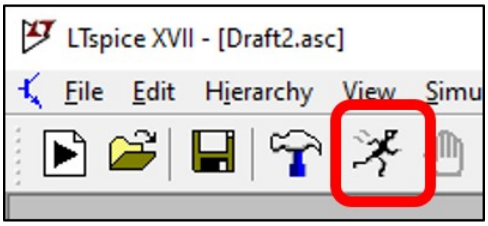

Figure 36: LTspice run simulation button.

You will be presented with a window containing the simulation results as shown in Figure 37. The window contains all the node voltages relative to ground (0 V) and the currents flowing through each component. You can now see why it is desirable to change component names in more complex circuits.
Note: The value in brackets of a voltage is the node name. You can see these names at the bottom left of the*LTspice* window when you hover the mouse cursor over a node.
Note: The current polarities in your circuit might differ. This will be discussed later.

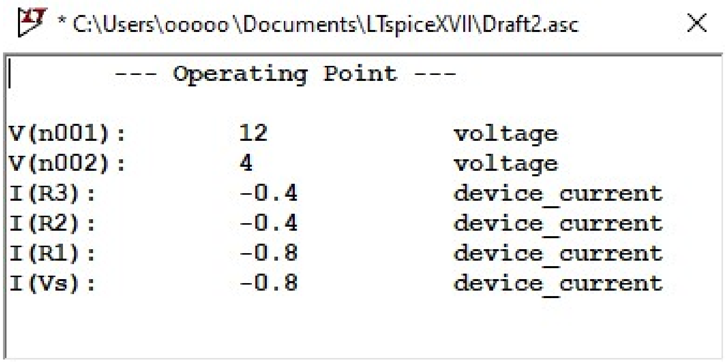

Figure 37: LTspice DC operating point simulation results.

5. Close the simulation result window (Figure 37) by clicking the X at the top right. You will now have the option to click on a node in your circuit to display the voltage at that node. Click on all the nodes to display their voltages. Add an additional voltage indicator above *R3* as indicated in Figure 38. Your schematic should look like Figure 38.

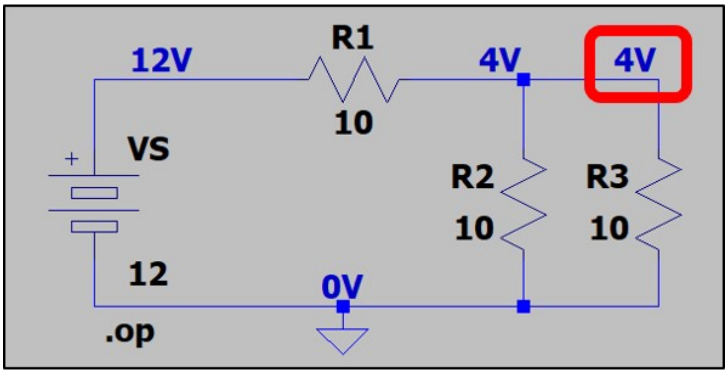

Figure 38: LTspice example with node voltages indicated.

6. To display a current on the schematic, right-click on the 4 V label indicated in Figure 38 to open the window shown in Figure 39. Clear the indicated text box.
   Select the current you want displayed by selecting it from the list. In this case, since the 4 V label is in the branch containing *R3* (see Figure 38), select the current through *R3* by clicking *I(R3)*. *I(R3)* will appear in the text box as shown in Figure 39.

   Tip: You can enter more complicated expressions into the textbox of Figure 39. As an example, you can display the power dissipated in *R3* by entering *V(n002)*I(R3)*.
   Note: For most simulation types (excluding DC operating point) you can add a current measurement by clicking on the body of a component. Similarly, power dissipation can be measured by pressing Alt and clicking on the component.

7. Click OK to close the window in Figure 39. You will now see the current through *R3* as shown in Figure 40 (The polarity of the current in your circuit might differ).

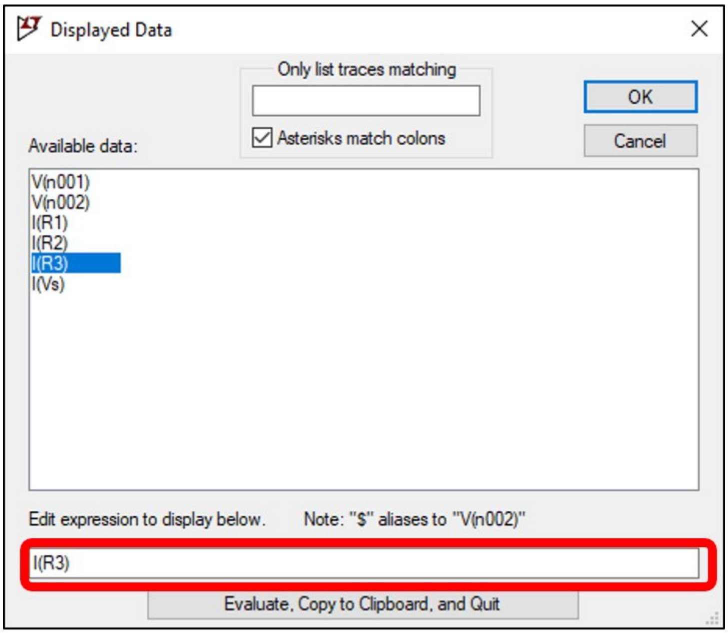

Figure 39: LTspice Displayed Data window.

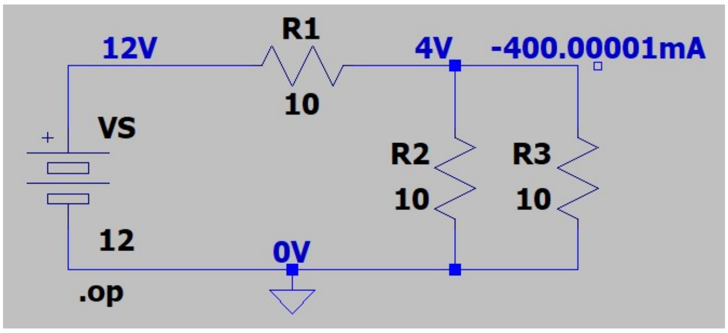

Figure 40: LTspice schematic with node voltages and R3 current.

8. Looking at the circuit layout and the polarity of the voltage source VS in Figure 40, it is clear that the current will flow from top to bottom through *R3*. So why is the current value negative? We can find the answer to this question by navigating to *View* → *SPICE Netlist* as shown in Figure 41.

   The *SPICE Netlist* window in Figure 42 will be displayed. Looking at the row for *R3*, we can see it is connected from node 0 (ground) to node *N002*, and has a value of 10 Ω. This means that the resistor *R3* has its positive reference connected to ground, and its negative reference to node *N002* (the 4 V node). The simulation is assuming that we chose the current polarity as indicated in Figure 43, and hence the negative current value we observed (current is flowing down through *R3*, but it was assumed that current flowing up is positive).

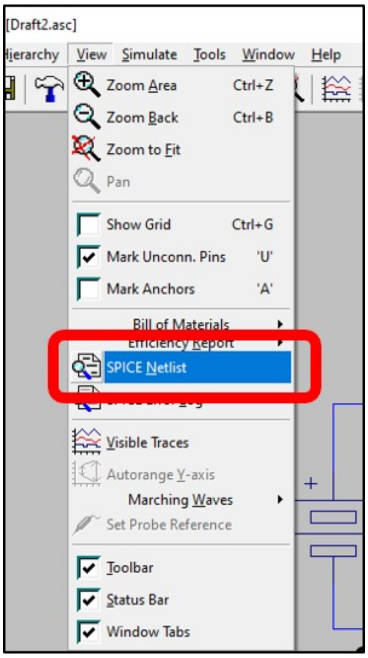

Figure 41: Drop-down menu to navigate to SPICE Netlist.

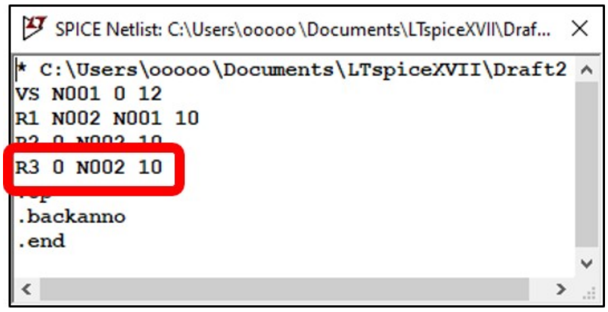

Figure 42: Example of SPICE Netlist window.

The problem can be fixed by flipping *R3* so its positive terminal is connected to node *N002*. Alternatively, you can interpret the results taking note of the assumed current direction.
In Figure 44, the simulation was repeated after flipping resistor *R3*. This was done to show that flipping the resistor gives the expected results. You can try this for yourself and confirm that the resistor is connected correctly by looking at the netlist as in Figure 42.

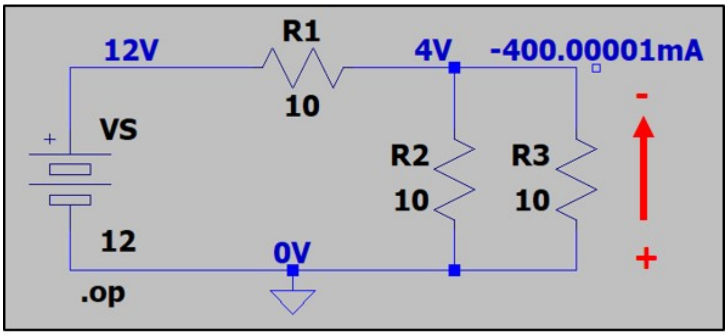

Figure 43: LTspice schematic with assumed R3 current polarity.

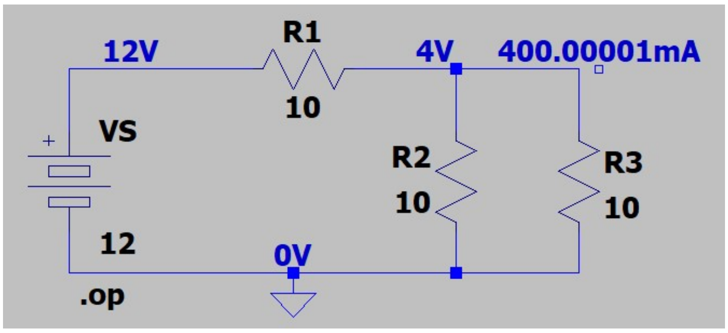

Figure 44: LTspice schematic with flipped resistor R3.

2.4.2 DC Sweep

We will now look at a DC sweep simulation. This type of simulation sweeps a value, such as the input voltage, through a range of values and measures the change in another set of values. As an example, we want to measure the current through *R3* and the voltage across *R1* as *VS* is changed from 0 V to 12 V in 0.5 V increments. The simulator will do a DC operating point simulation for each value of *VS*, and record all the node voltages and loop currents. We can then investigate the effect of changing component parameters.

1. The first step for setting up the simulation is to edit the simulation command. The process is similar to what was done for the DC operating point simulation. Go to *Simulate* → *Edit Simulation Command* (see Figure 32 to open the *Edit Simulation Command* window (see Figure 33). Navigate to the DC sweep tab to set up a DC sweep simulation.

2. Fill in the text boxes to set up the following simulation that was also described earlier: Sweep the VS source, linearly, from 0 V to 12 V in 0.5 V increments. The *Edit Simulation Command* window should look like Figure 45.

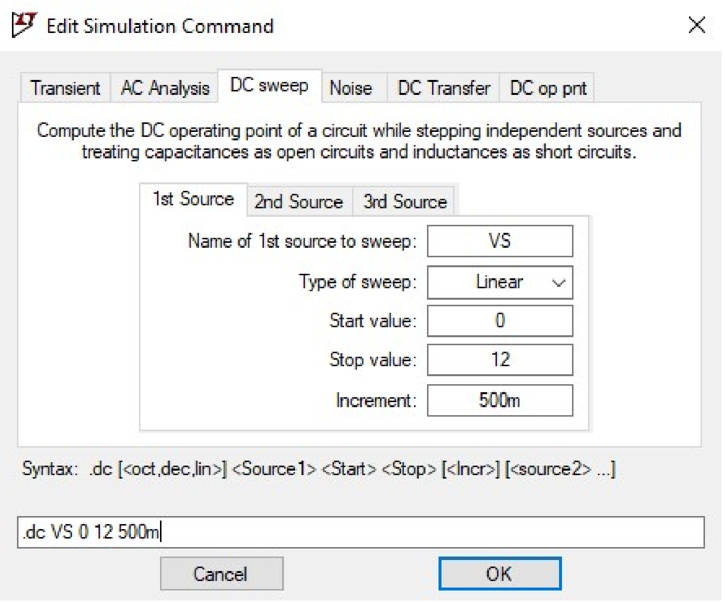

Figure 45: LTspice DC sweep simulation setup.

Note: You can see that if you give components proper names when building your circuit, setting up the simulation is much easier and you are less likely to make a mistake.

3. Click *OK* to close the *Edit Simulation Command* window. You can then place the simulation command text anywhere on the schematic. Your schematic should look like Figure 46.

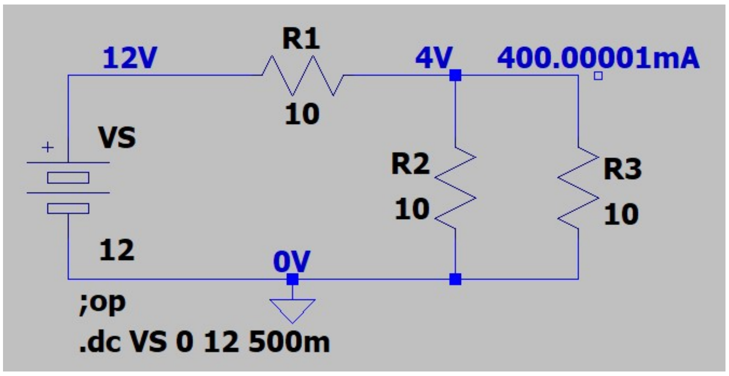

Figure 46: LTspice schematic with DC sweep simulation command.

Note: You will see that the previous simulation command is still present and has changed from ".op" to ";op". This indicates that the DC operating point simulation is no longer the active simulation. You can remove the old simulation command if you desire.

4. Click on the *Run* button (Figure 36) to run the DC sweep simulation. A blank graph window will open in the *LTspice* window as shown in Figure 47.

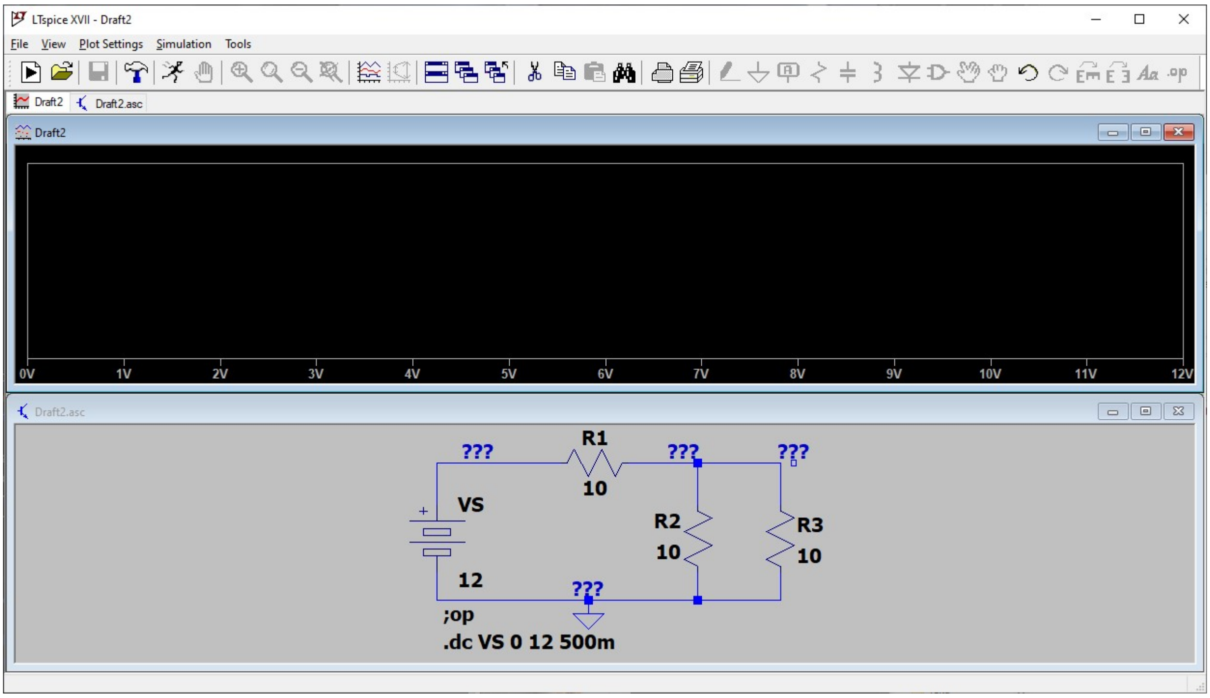

Figure 47: LTspice window after DC sweep simulation.

5. To add a trace (measurement) to the graph, you need to click on the node you want to measure. Click on the node at the positive side of *VS* as indicated by the red cross in Figure 47. A trace representing the voltage of *VS* will appear in the graph window as shown in Figure 48.
   Tip: The names of the traces are shown at the top of the graph window in the colour of the trace. In Figure 48, the trace is called *V(n001)*, where *N001* is the node name. In this circuit, the voltage of node *N001* is the same as the voltage of *VS*.
   Tip: You can delete a trace by pressing *Delete* and clicking on the trace name.
   Note: Because we are sweeping and measuring *VS*, the X (sweep value) and Y (measured value) values of the graph will be exactly the same. This confirms that the simulation is working as expected.

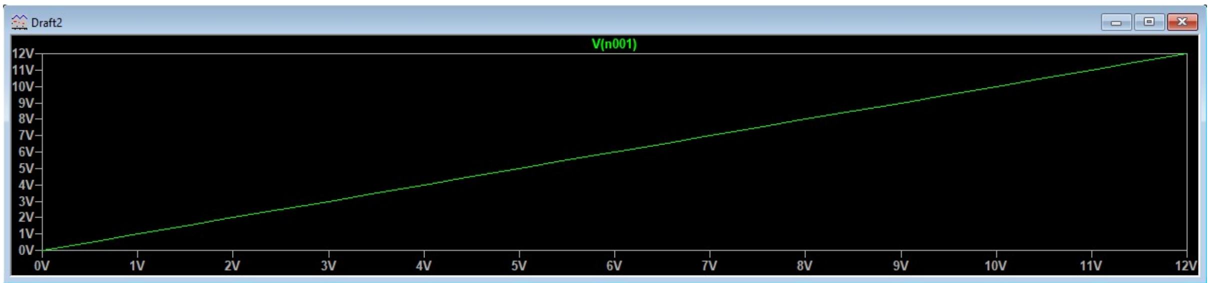

Figure 48: LTspice graph window with VS measurement (node n001).

6. To measure the current through *R3*, click on resistor *R3*. A trace should appear in the graph window named *I(R3)* as shown in Figure 49. You will see that the voltage and current values of the Y axis are displayed on the left- and right-hand sides respectively.

   Note: You can also add a power dissipation measurement by pressing *Alt* and clicking on the component.

7. The last measurement we want to make is the voltage across *R1*. To do this, we need to find the voltage difference between the nodes on either sides of resistor *R1*, as indicated by the green and

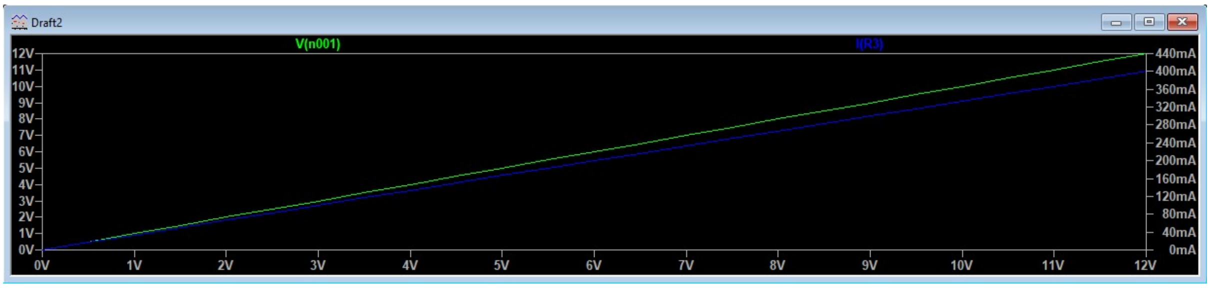

Figure 49: LTspice graph window with current measurement of R3.

blue crosses in Figure 50. Since the voltage polarity is not given, the green side is assumed to be the positive voltage terminal.

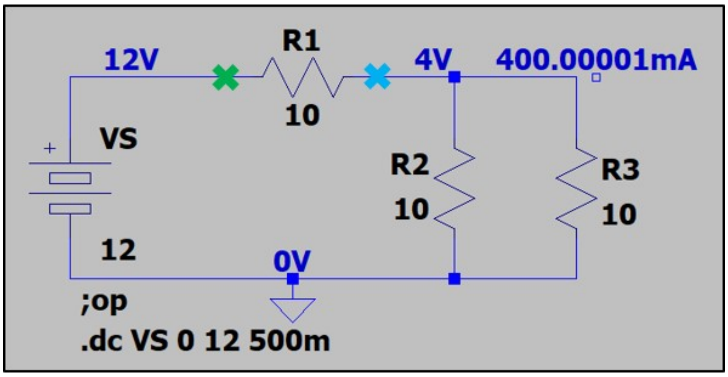

Figure 50: LTspice schematic indicating R1 voltage nodes.

Find the node names at the green and blue crosses in Figure 50 by hovering the cursor over the nodes and observing the node names at the bottom left corner of the window. Remember the names (*N001* and *N002* in the example) and which one is the positive terminal (*N001* in the example).

8. To add the trace representing the voltage across *R1* to the graph, right click anywhere on the graph and click on *Add Traces* as shown in Figure 51. The *Add Traces to Plot* window shown in Figure 52 will be displayed.

9. In the *Add Traces to Plot* window shown in Figure 52 , click on the voltage at the positive node identified before (*V(n001)* in this example) to add the voltage to the expression box indicated in the red box. Then type a *minus* (-), and click the voltage at the negative node (*V(n002)* in the example). Your expression should look like the one in Figure 52. Click *OK* to close the window and add the trace to the graph. A new trace representing the voltage across *R1* will appear in the graph window as shown in Figure 53 (red trace).

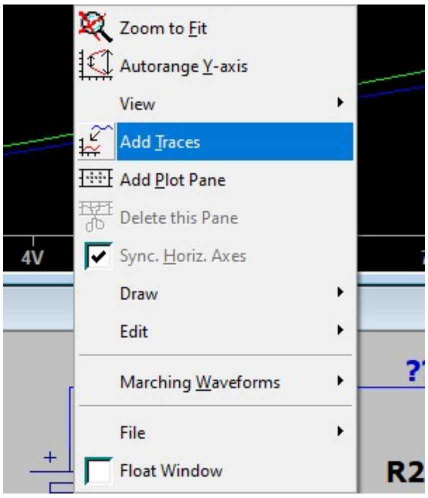

Figure 51: LTspice Add Traces menu.

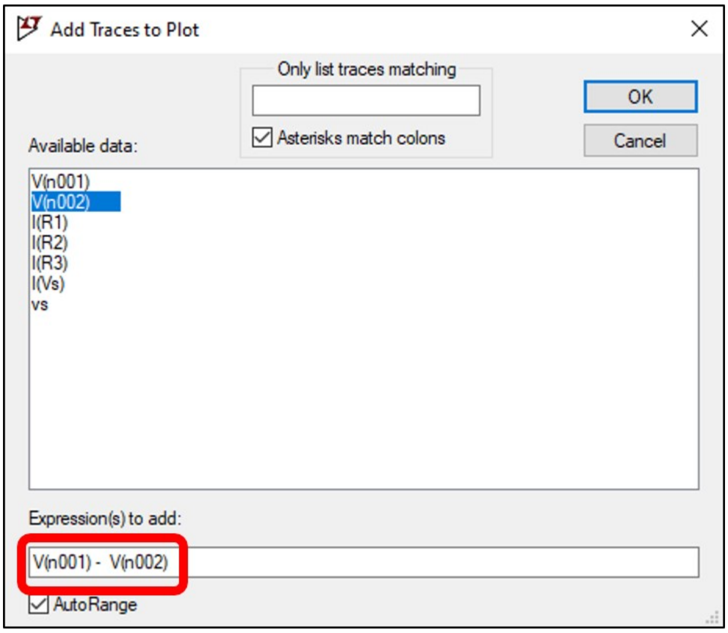

Figure 52: LTspice Add Traces to Plot window.

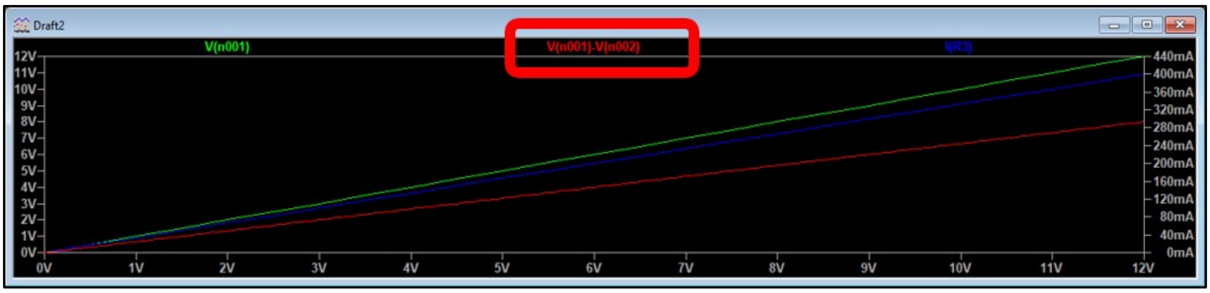

Figure 53: LTspice graph window with voltage measurement across R1.

10. Looking at Figure 53, what is the voltage across *R1* when *VS* is 4.3 V? You can see that it is difficult to take accurate measurements from the graphs. We can use cursors to get very accurate readings of the X and Y axes.
    Add a cursor to the *R1* voltage trace by clicking on the trace name indicated in Figure 53. A cursor will appear on the selected trace (red trace in this example) as well as a *Cursor* window displaying the cursor values as indicated by the red block of Figure 54.
    Click and drag the cursor to the point on the X axis where *VS* is 4.3 V (use the *Cursor* window to accurately align the cursor). Using a cursor, you should find that the voltage across *R1* is 2.87 V when *VS* is 4.3 V.

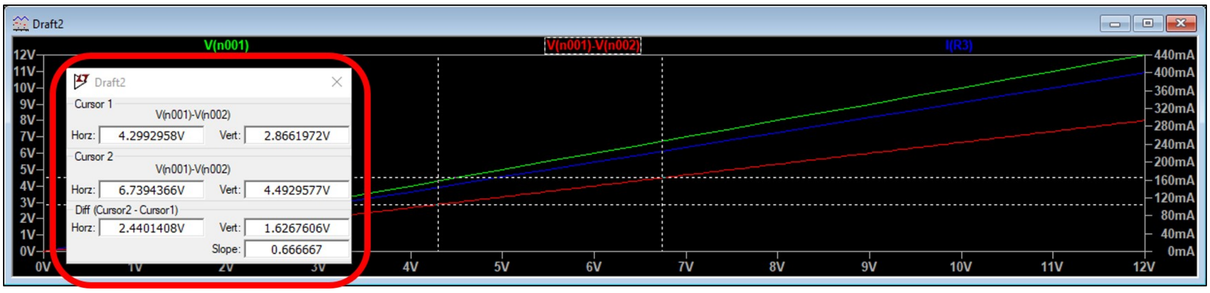

Figure 54: LTspice graph window with cursors.

11. Click on the trace name again to add a second cursor to the graph of Figure 54. Your graph window should now look similar to Figure 54. A second cursor allows additional measurements such as the difference between two points on the trace and the slope of the trace.

2.5 Resistance sweep

Unlike voltage and current, resistance sweeping is not possible to do in the DC Sweep section under Edit Simulation Command. The convenient way to sweep resistance is through the spice directive. The same process on how to setup a voltage sweep or current sweep using the spice directive, the resistor to be varied must be enclosed in curly braces.

2.5.1 Resistance sweep using incremental sweeping

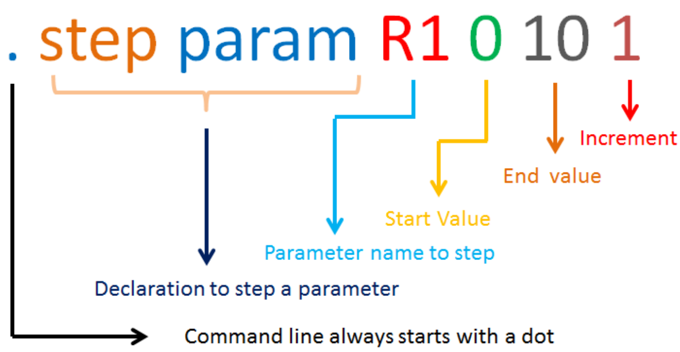

Figure 55: LTspice incremental sweeping parameters

The first method in how to sweep resistance in LTSpice is by using incremental sweeping. Meaning, you are going to sweep a resistance value from a specific starting value towards a specified end value with a fixed interval as in Figure 55. This is also called linear sweeping. In the circuit in Figure 56 we want to vary the value of R1 and monitor the V_{out} level.

1. On the spice directive window, enter the step command as shown below in Figure 61. Press keyboard letter "T" to show the spice directive. Do not forget to tick the "Spice directive".

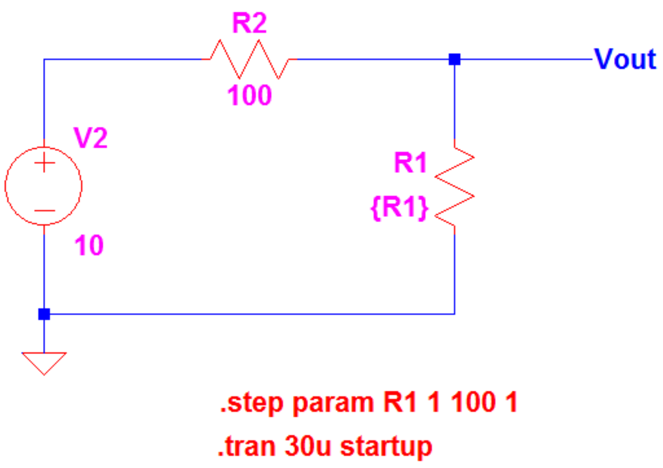

Figure 56: LTspice incremental sweeping circuit

Otherwise, your inputs will be treated as comment only and it will not run. In this setup, R1 value is varied from 1 Ω to 100 Ω with an increment of 1 Ω. Another way to show the spice directive is by clicking the icon in Figure 57.

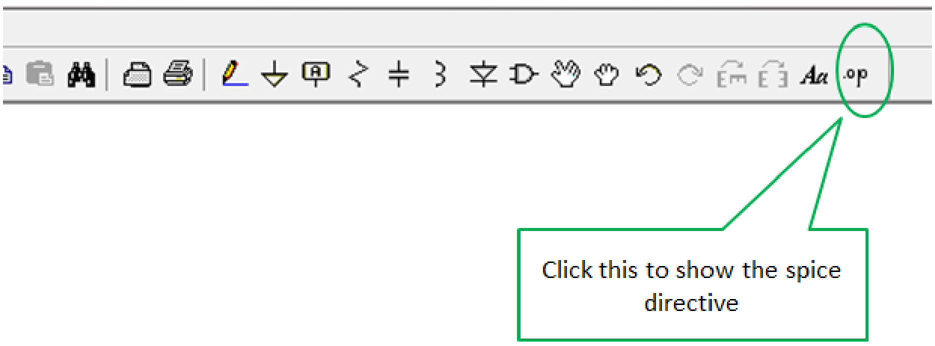

Figure 57: LTspice incremental sweeping spice directive

2. Setup the transient command as shown in Figure 58. Click on "Simulate" icon bar then "Edit Simulation Cmd" to show the Transient section. In this example Transient command is being used. However, it does not mean that you cannot use the other commands. The simulation results start when the box for Start external DC supply voltage at 0V is ticked. When the "Start external DC supply voltages at 0V" is ticked, the simulation will start at zero and the ramping or rising of the waveforms will be those shown in Figure 59. Otherwise, when the "Start external DC supply voltage at 0V" is unticked, the waveform will only show the steady state as seen in Figure 60. This is not the only method to sweep resistance in LTspice.

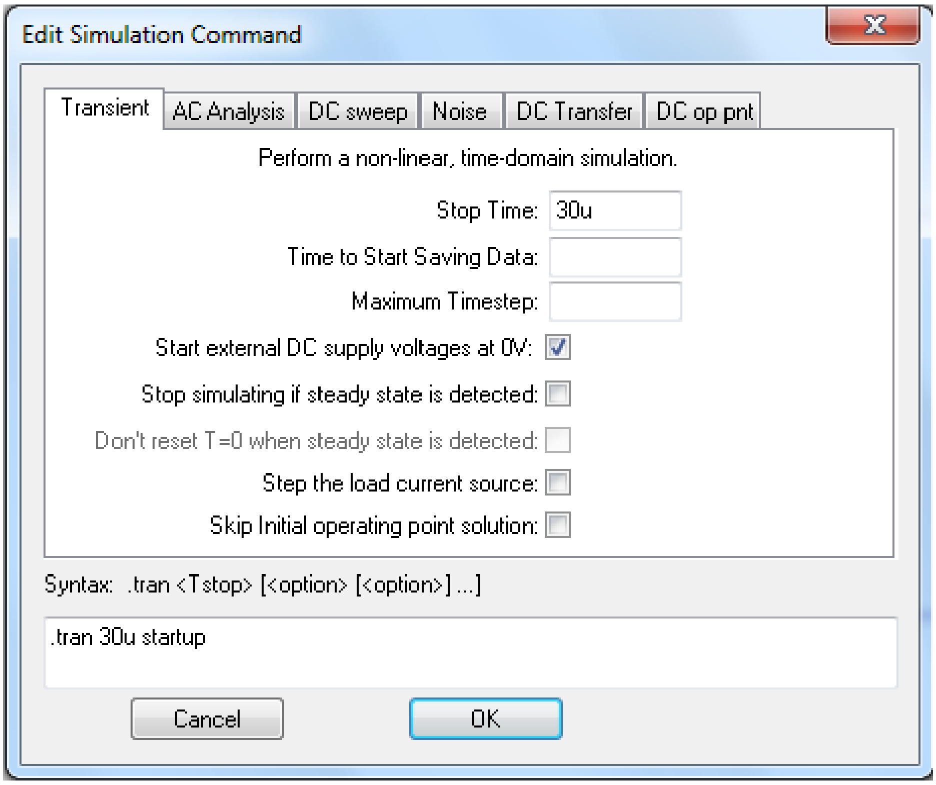

Figure 58: Edit simulation command

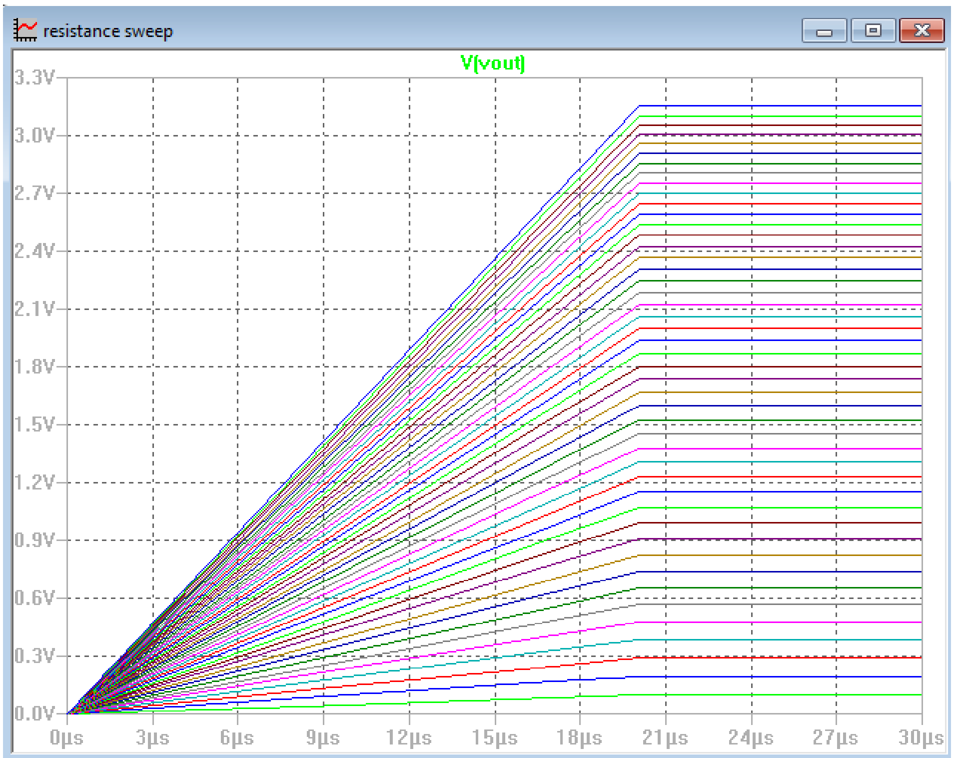

Figure 59: Resistance sweep waveforms

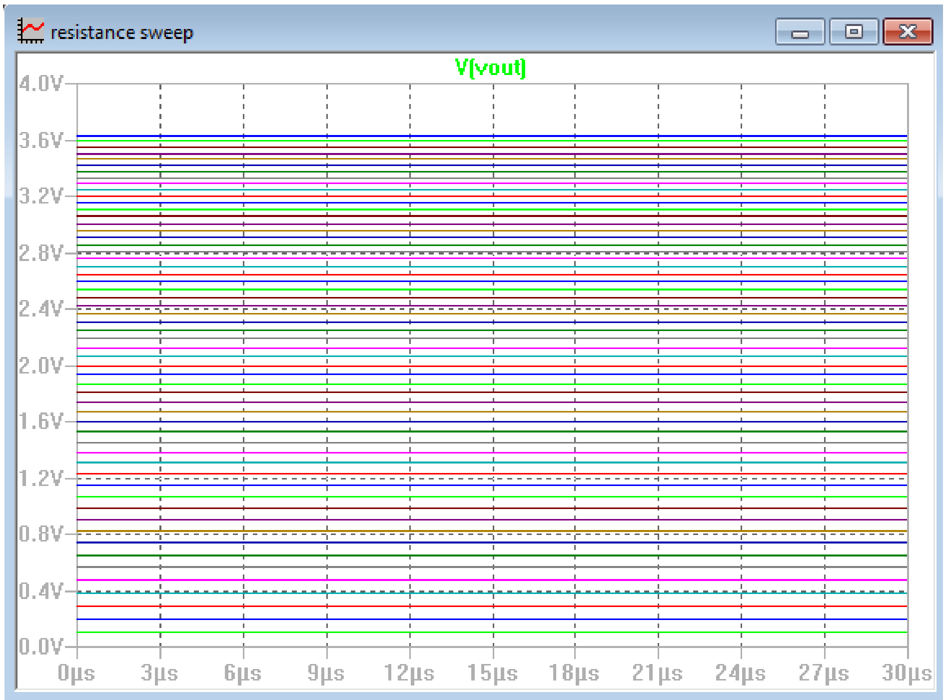

Figure 60: Resistance sweep steady-state waveforms

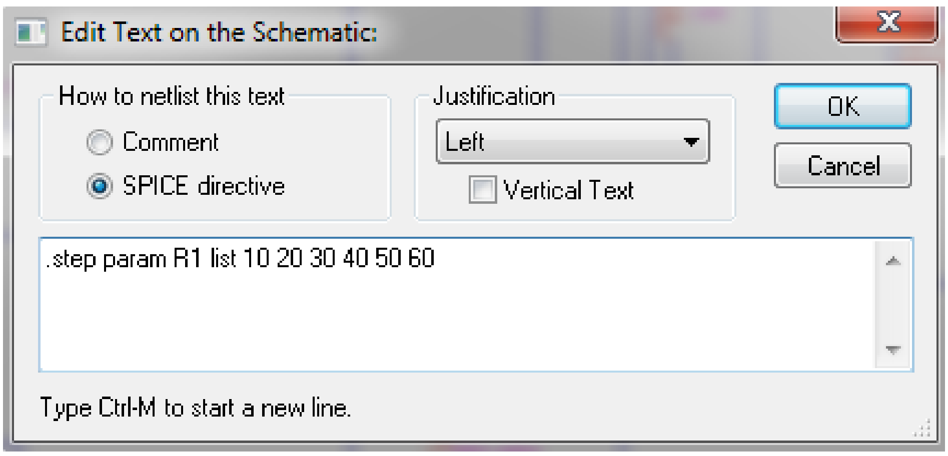

Figure 61: LTspice incremental sweeping directive window

2.6 Dependent sources in LTSpice

In LTSpice, there are two main types of dependent sources: Voltage Controlled Sources (VCS) and Current Controlled Sources (CCS).

Voltage Controlled Sources:

1. Voltage Controlled Voltage Source (VCVS): The output voltage of this source is proportional to the input voltage across a different part of the circuit.

2. Voltage Controlled Current Source (VCCS): The output current of this source is proportional to the input voltage across a different part of the circuit.

Current Controlled Sources:

1. Current Controlled Voltage Source (CCVS): The output voltage of this source is proportional to the input current through a different part of the circuit.

2. Current Controlled Current Source (CCCS): The output current of this source is proportional to the input current through a different part of the circuit.

These dependent sources are useful for modeling and simulating circuits that contain feedback or control elements. They can be used to implement various control and signal processing functions, such as amplifiers, filters, and oscillators.

2.6.1 Voltage Controlled Voltage Source

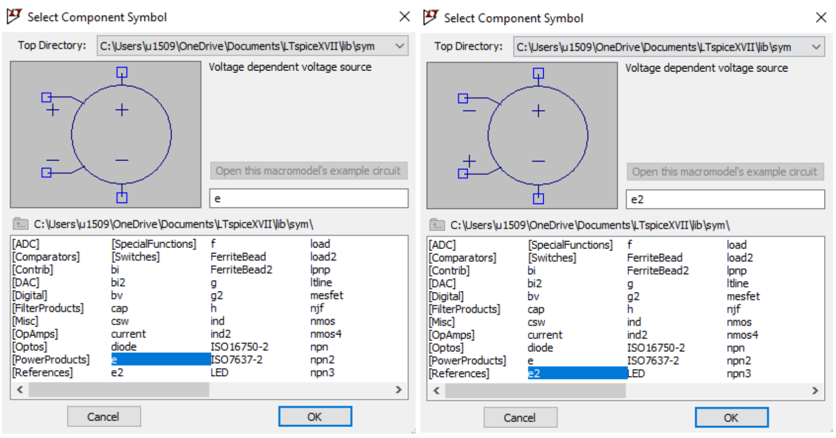

Figure 62: LTSpice component selection of a voltage dependent voltage source

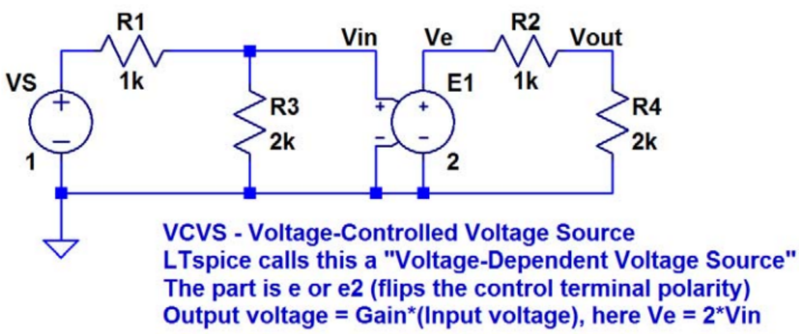

Figure 63: LTSpice circuit example for connecting a VDVS

2.6.2 Voltage Controlled Current Source

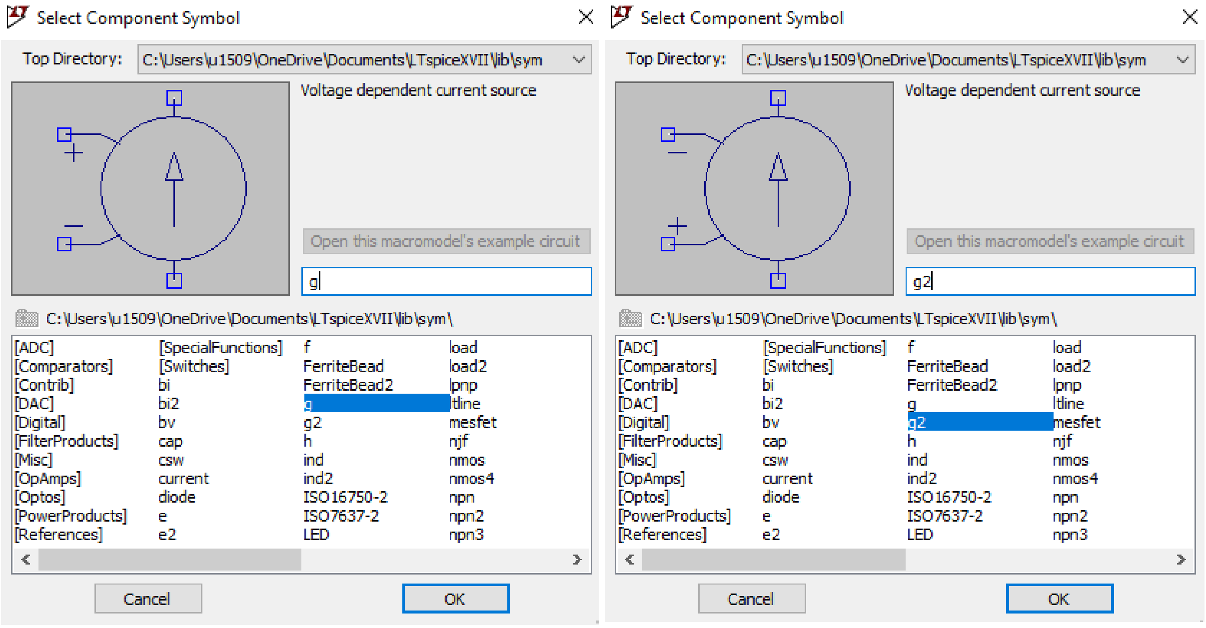

Figure 64: LTSpice component selection of a voltage dependent current source

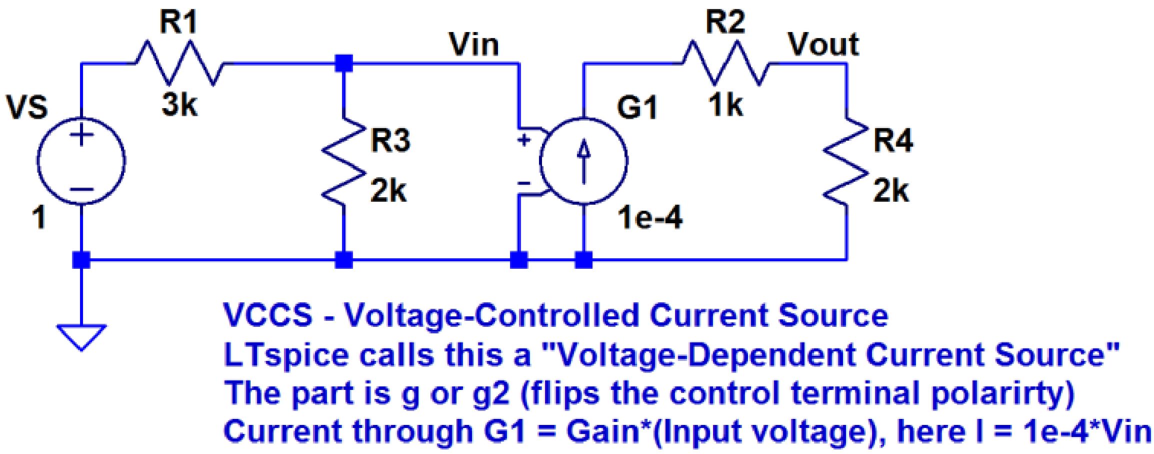

Figure 65: LTSpice circuit example for connecting a VCCS

2.6.3 Current Controlled Voltage Source

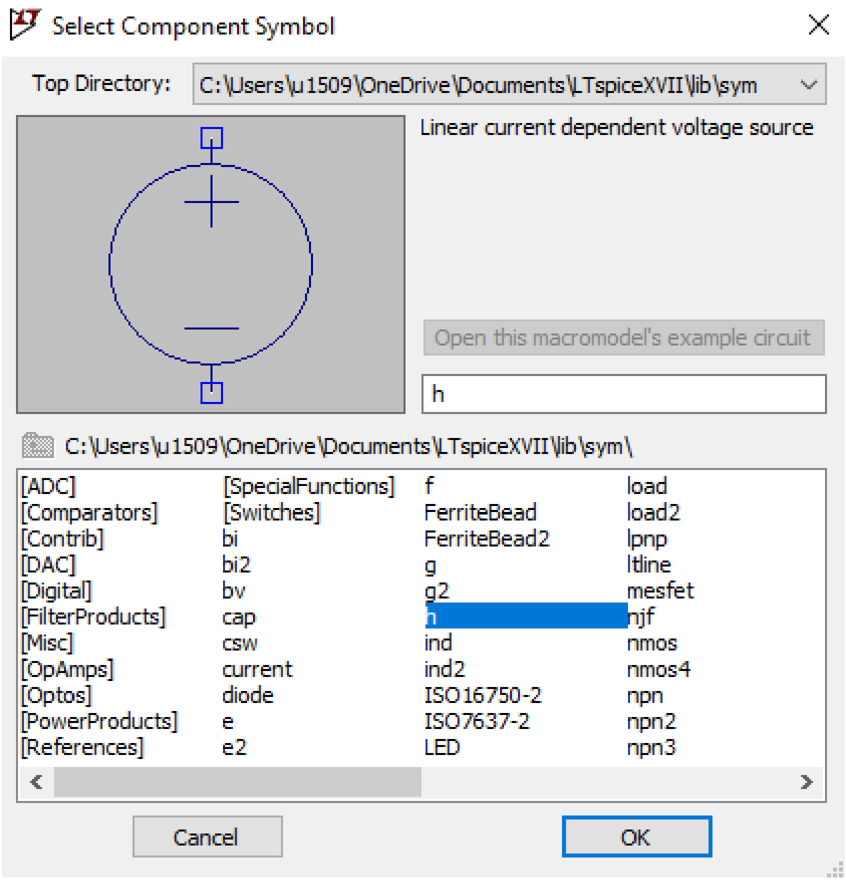

Figure 66: LTSpice component selection of a linear current dependent voltage source

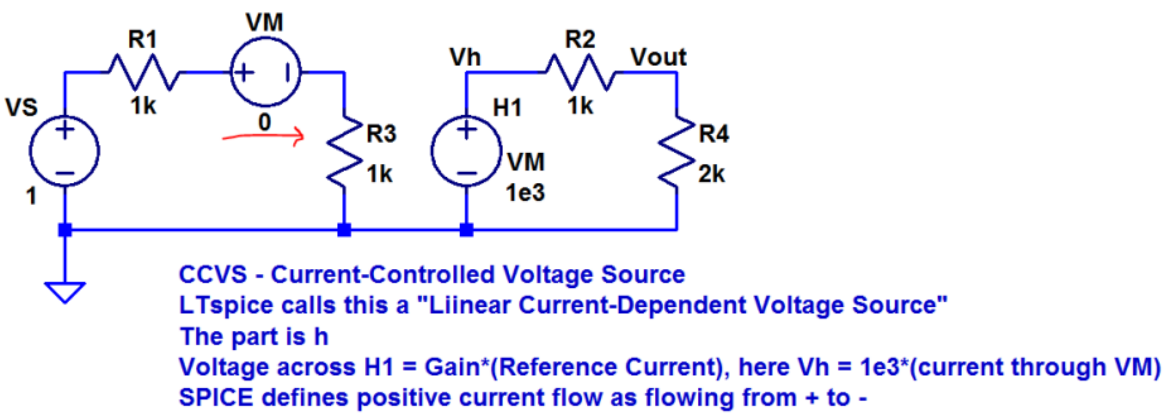

Figure 67: LTSpice circuit example for connecting a CCVS

2.6.4 Current Controlled Current Source

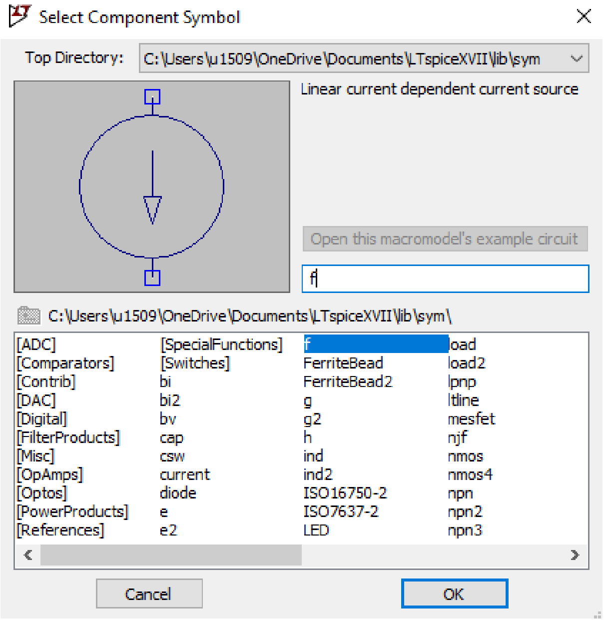

Figure 68: LTSpice component selection of a linear current dependent current source

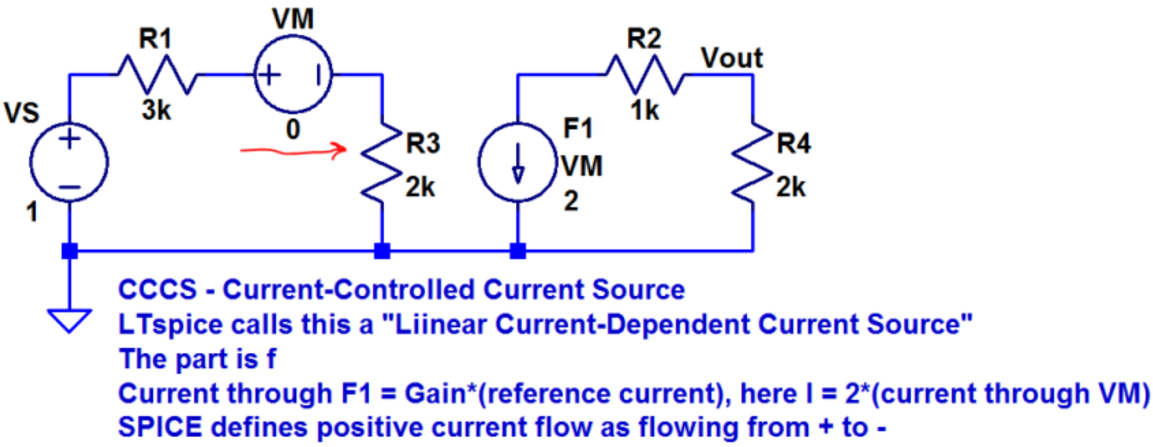

Figure 69: LTSpice circuit example for connecting a CCCS

2.7 Creating a current dependent current source in LTSpice

This section introduces a few examples of creating a current dependent current source in LTSpice. The method uses the arbitrary behaviour current source, or "bi" (Fig. 70) default library component. It's very important to understand the current direction convention LTspice is using for various components, as will be demonstrated below for other current sources, voltage sources, components with only two terminals, and components with more than two terminals. Understand that in all the given examples, even though all the independent currents are mirrored directly by the dependent current source, it's also possible to adjust them with any arbitrary mathematical function LTspice recognizes, hence the component being named "Arbitrary Behavior Current Source". For a full list of valid mathematical operators and functions search for "bi" in the help topics inside LTspice.

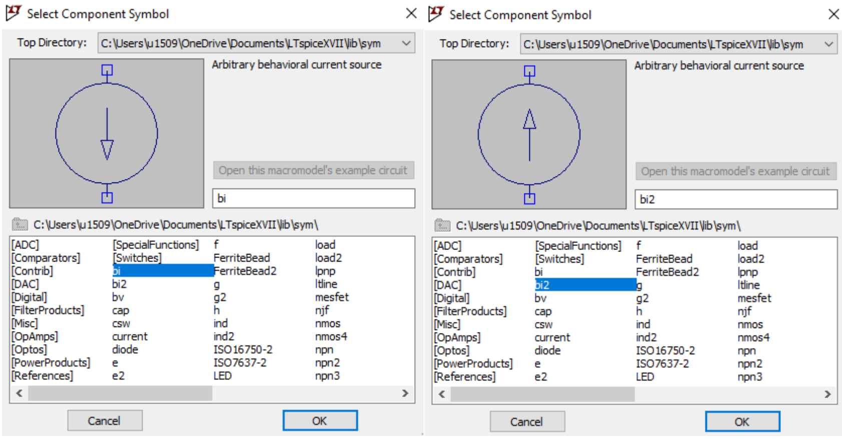

Figure 70: LTSpice component selection of an arbitrary behaviour current source

2.7.1 A current source dependent on another current source

This is very straightforward, the convention here is that the direction of current flow matches the arrow indicators between the dependent current source and the independent current source. Here, an independent 1mA DC current source, I_1, is connected to a 1 kOhm load, and a dependent current source, B_1, is set to mirror I_1's current exactly ($I = I(I_1)$) through a larger 10 kOhm load. Very easy, and works as expected with 1V at Vin and 10V at Vout. This is shown in Fig. 71. If the independent source is flipped 180 degrees so that it's driving its 1mA current the opposite direction as seen in Fig. 72, the dependent current source circuit is unaffected because current flowing out of the independent current source arrow flows out of the dependent current source arrow. Again, this works as expected, this time with -1V at Vin but still 10V at Vout.

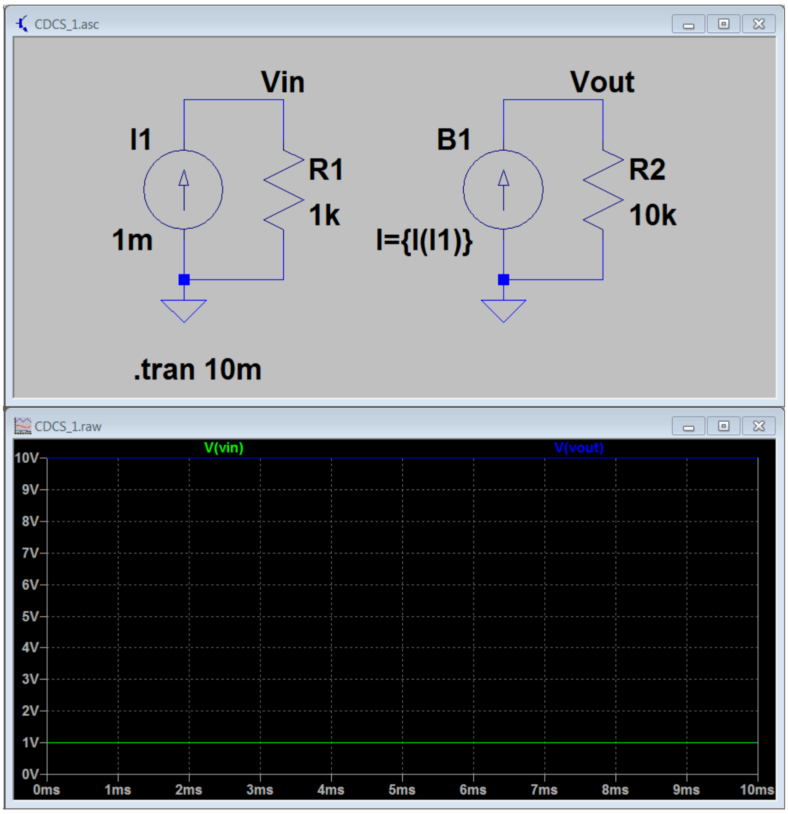

Figure 71: A current source dependent on another current source "bi2" (Fig. 70)

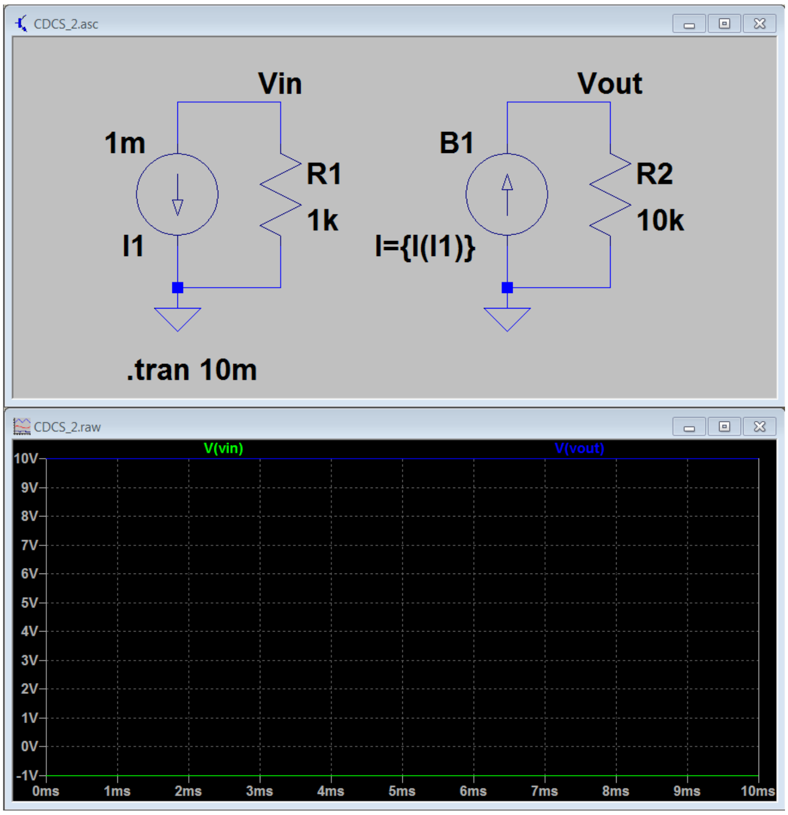

Figure 72: A current source dependent on another current source "bi" (Fig. 70)

2.7.2 A current source dependent on the current of a voltage source

The convention in this configuration is that positive current flows in to the positive terminal. It isn't difficult to make an incorrect assumption based on this. In Fig. 73, the convention makes the dependent source's current flow opposite the direction its arrow is pointing. This might be counter-intuitive at first, but it's easy to fix the current direction by simply subtracting in the dependent current source equation ($I = -I(V1)$ instead of $I = I(V1)$), as shown in Fig. 74.

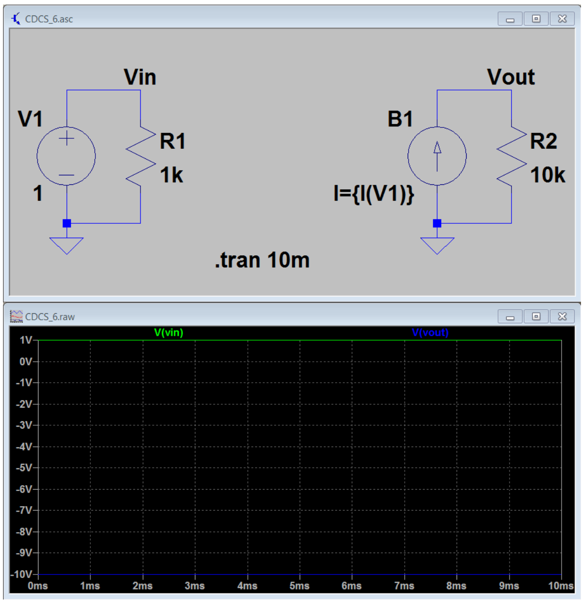

Figure 73: A current source dependent on the current of a voltage source with incorrect polarity

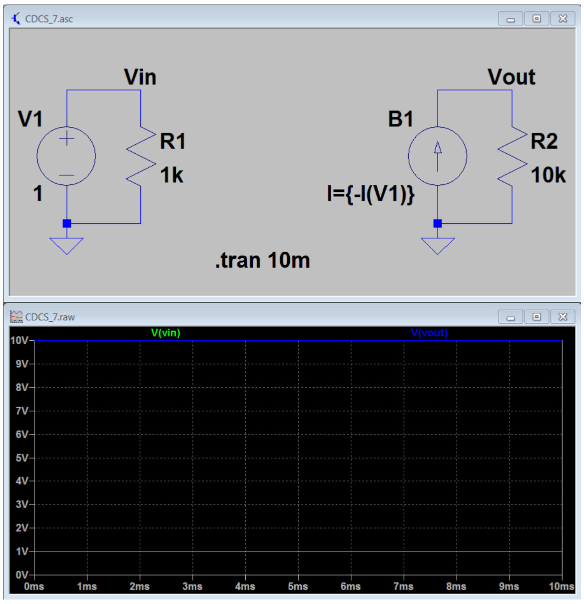

Figure 74: A current source dependent on the current of a voltage source with correct polarity

2.7.3 A current source dependent on the current of a two terminal component

For better or worse, LTspice treats two terminal components differently than components with three or more terminals. For components like resistors, capacitors, and inductors this can make it difficult to know what the current convention is prior to actually running a simulation (especially if components have been mirrored and/or rotated). Observe and note that the dependent current equation is the same in both Fig. 75 and Fig. 76 ($I = I(R1)$).

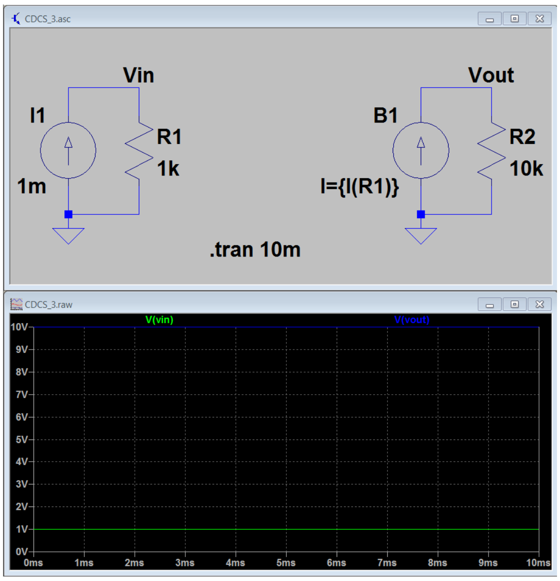

Figure 75: A current source dependent on the current of a two terminal component with unknown polarity

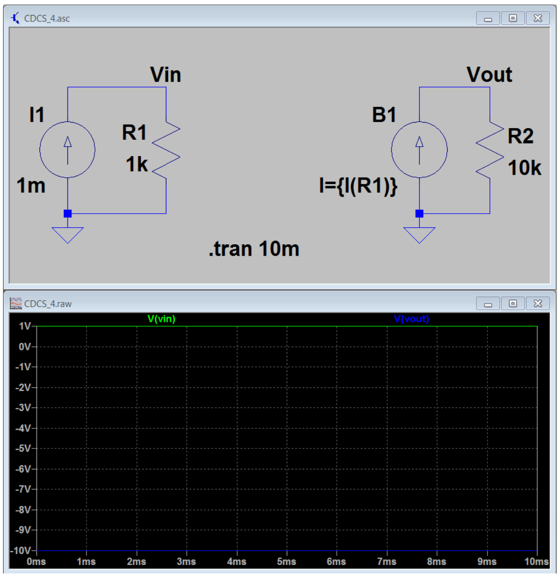

Figure 76: A current source dependent on the current of a two terminal component with unknown polarity

After running the simulation, a simple means of checking the current convention of a two terminal component is to hover over the component in the schematic window and look at the current probe graphic that appears. This can be seen in Fig. 77, where the positive direction for current is indicated by the arrow of the probe.

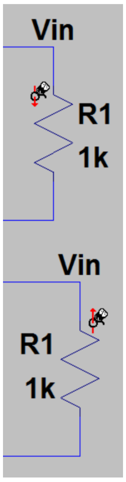

Figure 77: The polarity of Fig. 75 and Fig. 76

2.7.4 A current source dependent on the current of a three terminal component

For a component with three or more terminals the convention is that current flowing into a pin is positive while current flowing out of a pin is negative. Here are some examples with a simple NPN transistor circuit, where the convention as stated should make the base and collector currents positive and the emitter currents negative. Sure enough, this is true in Fig. 78 and Fig. 79.

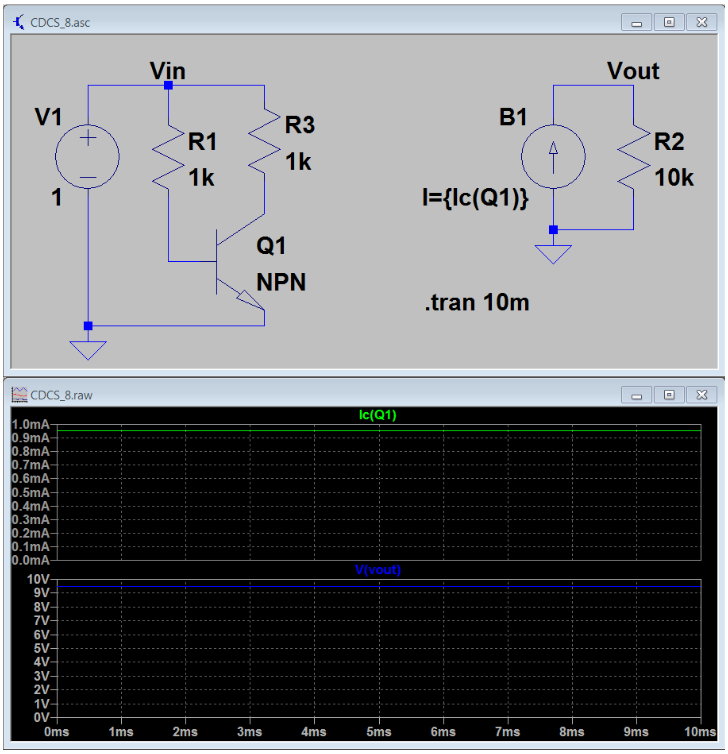

Figure 78: A current source dependent on the collector current of a three terminal component

Note the different dependent current equations ($I = I_c(Q1)$ in Fig. 78 vs. $I = I_e(Q1)$) in Fig. 79. Unlike probing a two-pin component, adding a current probe for a three or more terminal device requires clicking on the specific pin to make the distinction, as seen in Fig. 80. Note that the current probe graphic has now changed and indicates no inherent direction, as the previously mentioned positive in and negative out convention is used for each pin. Also, keep in mind that the dependent

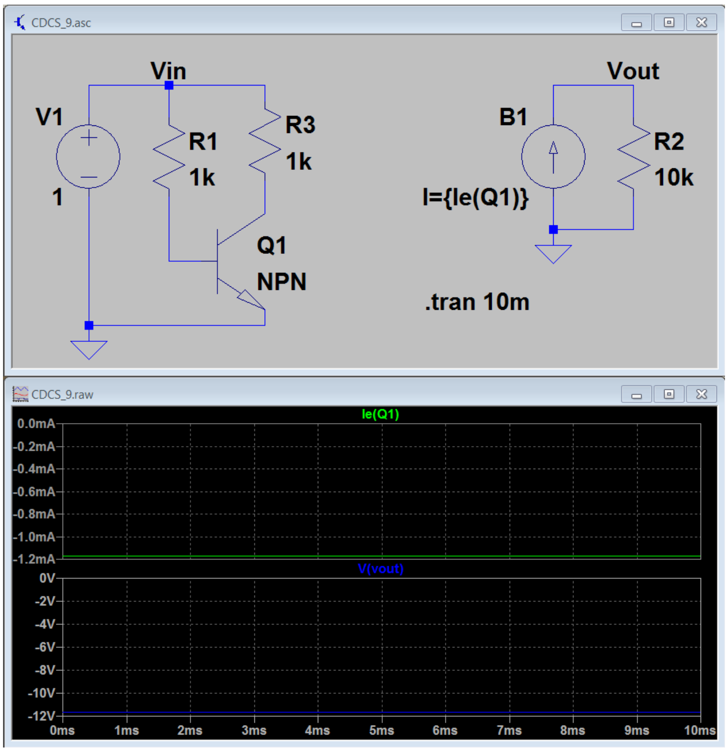

Figure 79: A current source dependent on the emitter current of a three terminal component

source equation is arbitrary, and therefore if the emitter current were desired to be made "positive", this could be done by subtracting it (as before with a voltage source current), or since it's known that the emitter current is equal and opposite to the sum of the collector and base currents ($I = I_c(Q1) + I_b(Q1)$), etc. This is shown in Fig. 81. The only limitation is mathematical imagination and the functions/operators provided in LTspice.

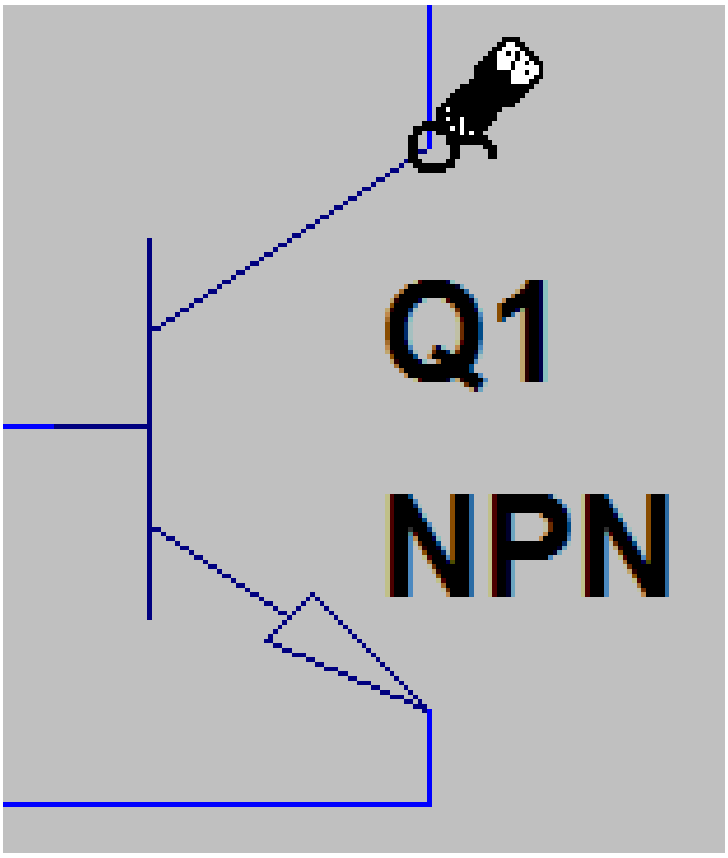

Figure 80: Probing a three terminal component dependent current source

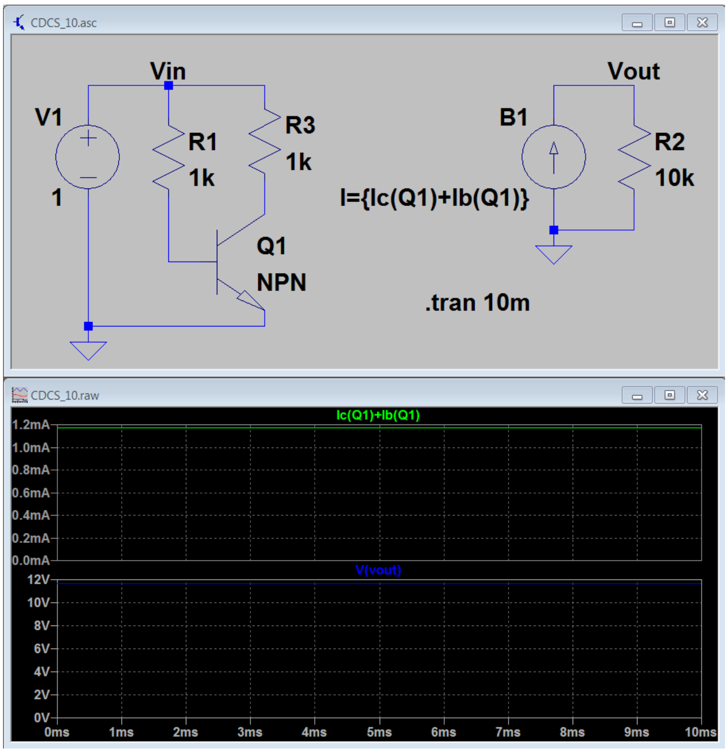

Figure 81: An alternative emitter equation for a current source dependent on the current of a transistor

3 Lab Environment

3.1 Building a Real Circuit

3.1.1 Protoboards/breadboards

Protoboards (or breadboards) are used to build prototype circuits. Component leads and wires are pushed into the holes in the protoboard to make electrical connections. You will build all your circuits on your protoboard for the EBN 111 practicals.
The holes are internally connected as shown in Figure 82 by the thick lines. It may be useful to think of each line as a node in your circuit. Note that the long connections at the top and bottom may differ with other protoboard models. An example of how resistors are connected in series and in parallel is shown in Figure 82.

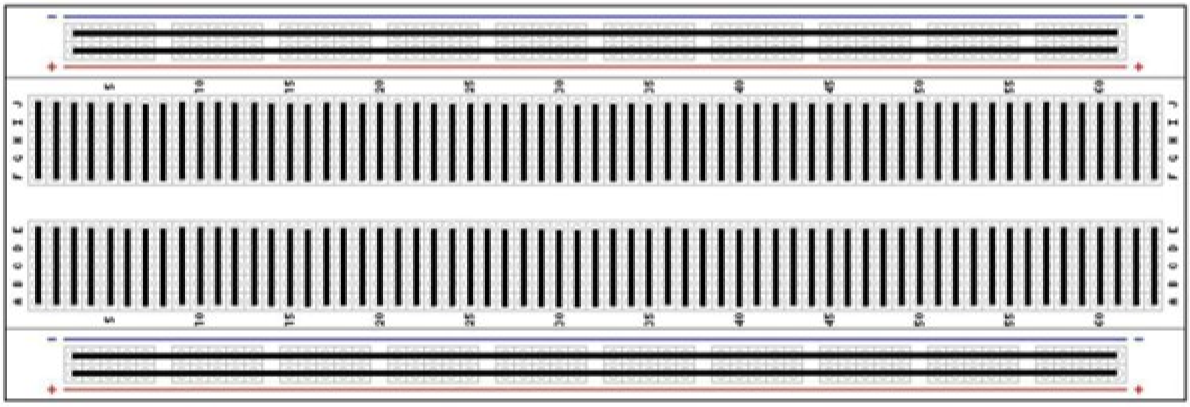

Figure 82: Internal connections of the protoboard. Thick lines indicate metallic connections (nodes).

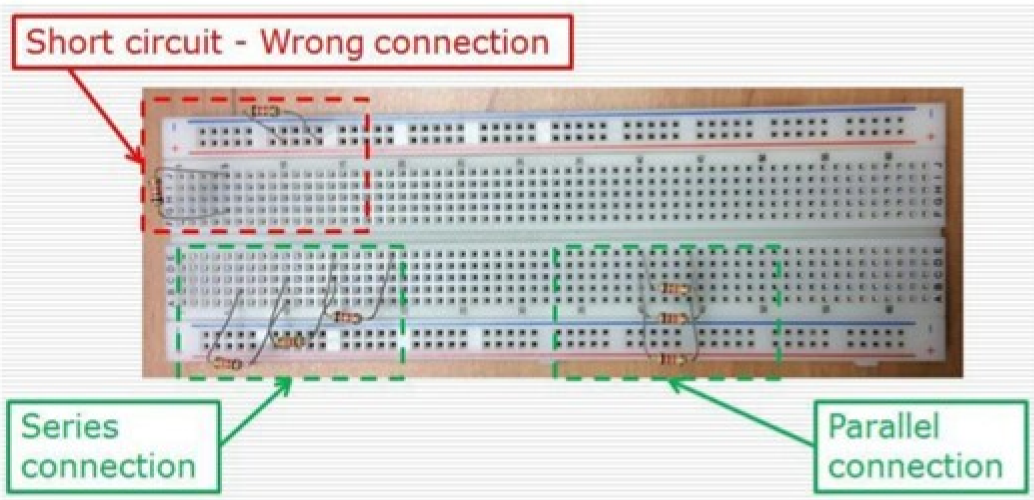

Figure 83: Example of resistors connected in series and parallel on the protoboard.

3.2 Lab Equipment

3.2.1 DC Power Suppy

A laboratory power supply (Figure 84) is used to supply electrical power to a circuit. Power supplies allow you to set the voltage that you need for your circuit and also allows you to limit the current (to prevent damage to the circuit if there are errors in the circuit). Do not use the power supply's earth connection (green connection) for your circuit's ground, rather use the COM connection (black connection). Banana plugs as in Figure 14 (a) are used to connect to the power supply. The other end of the cables are crocodile clips as in Figure 14 (b) that are connected to the circuit.

Figure 84: A laboratory power supply.

3.2.2 Multimeter

A digital multimeter (Figure 85) is an instrument capable of measuring voltage, resistance and other parameters in a circuit. The multimeter is connected using the same banana/crocodile clip cables as the power supply.

Figure 85: A digital multimeter.

3.2.3 Function Generator

A laboratory function generator is an instrument capable of generating voltages that vary with different waveforms. The function generator also allows you to set the amplitude and frequency of the generated signal. The function generators in the CAEC lab are built into the oscilloscopes (Figure 86).

3.2.4 Oscilloscope

An oscilloscope (Figure 86) is used to monitor varying voltages in a circuit. It plots a continuous graph with time on the x-axis and the measured voltage on the y-axis. Oscilloscopes also have the capability of measuring the frequency, amplitude and rise/fall time of voltage signals. Both the function generator and the oscilloscope are connected via BNC cables (Figure 14 (c)). The other side of a BNC cable is pair of crocodile clips. The red side is used to measure/supply the voltage and the black is connected to ground.

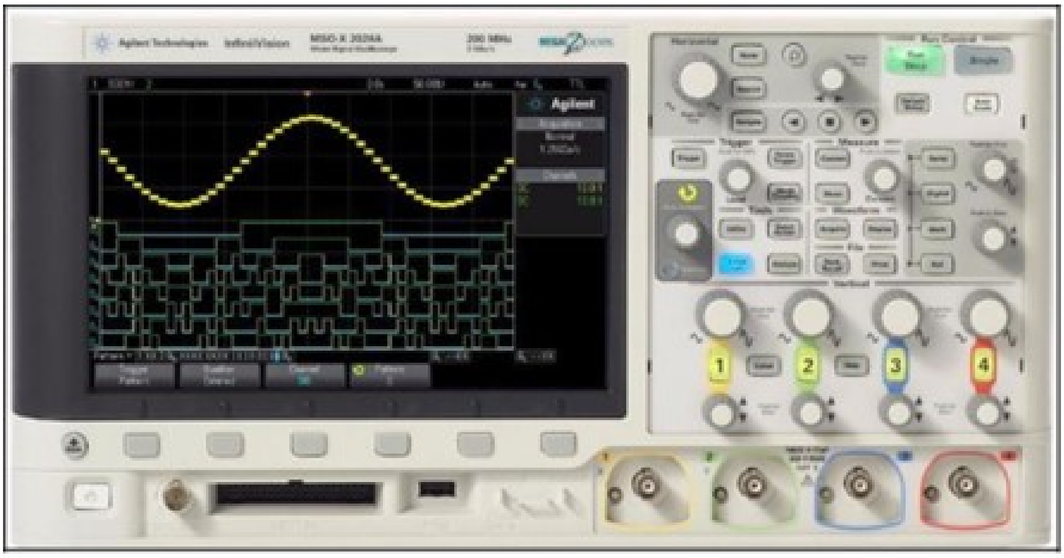

Figure 86: An oscilloscope with a built-in function generator.

3.3 Basic Experimental Setup

A basic experimental setup, using the circuit given in Figure 44 is given below.

3.3.1 DC Circuits

If a DC voltage is supplied to the circuit, then the following steps are performed:

1. The circuit is prepared on a breadboard using 3 10 Ω resistors, 2 in parallel and the 3rd in series as shown in Figure 86.

2. 3 wires are connected to important nodes. Red is connected to the positive terminal of the voltage source, black to the ground and orange to the node at which measurements must be taken.

3. The power supply as shown in Figure 88 is set up by connecting a cable to the COM port (ground) and the +20 V port. If a voltage of less than 6 V is required then the +6 V port can be used. Similarly, the -20 V port is used to supply a negative voltage.

4. To veiw the voltage output of the respective port, the correct button on the bottom left (under 'METER') must be selected).

5. To adjust the voltage, the knobs on the top right are turned. The precision of the power supply can sometimes be quite low, aim to get the voltage within 0.1 V of the desired value.

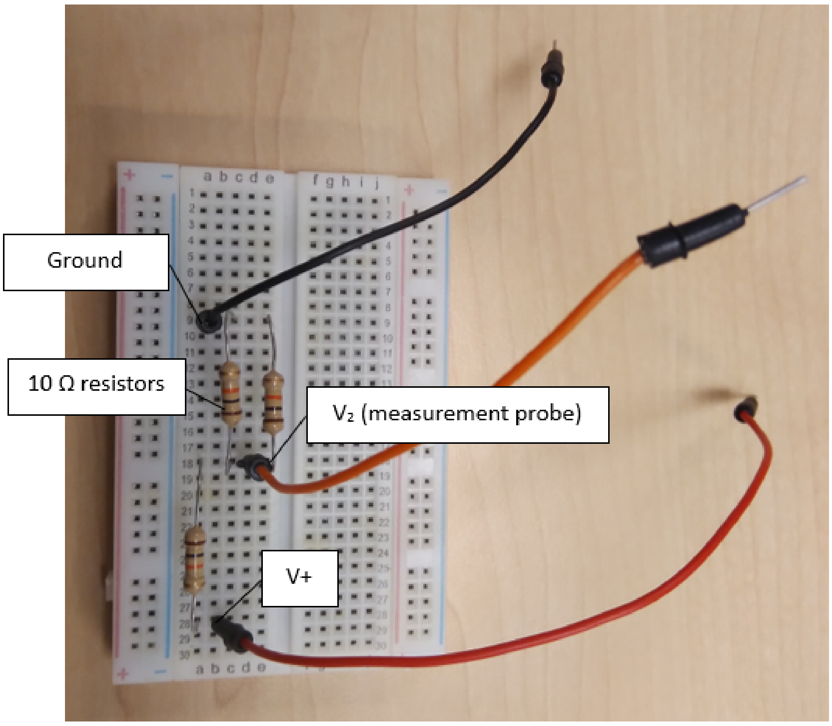

Figure 87: Circuit in Figure 36 built on a breadboard.

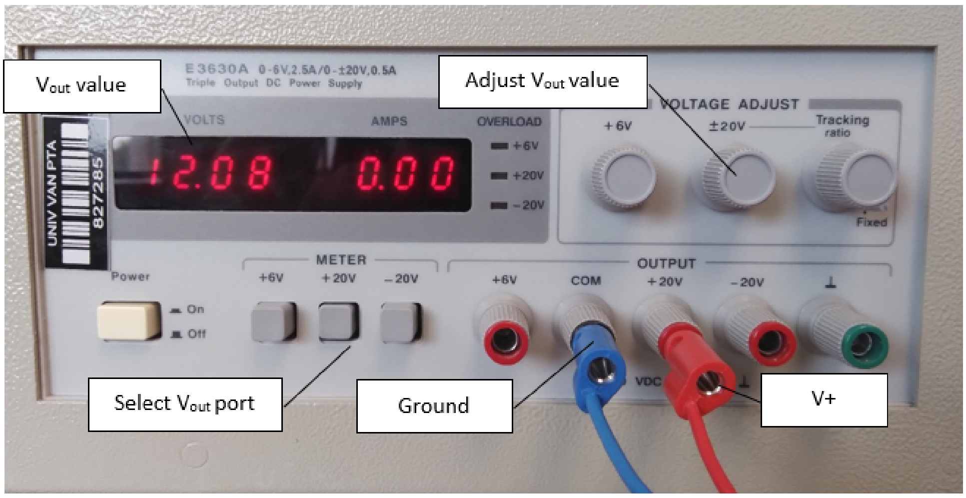

Figure 88: Connectors attached to the power supply for a 12 V source.

6. The multimeter is set up using the cables as shown in Figure 89. The other ports are used for measuring current and will not be used for practicals in EBN 111. The 'DCV' button is pressed to ensure that the measured voltage is displayed. The 'Ω' button can be used to measure the resistance between two points of a circuit.

7. The cables from the power supply and the multimeter are connected as shown in Figure 90. The ground of the multimeter must be connected to the same cable connected to the 'COM' port of the power supply.

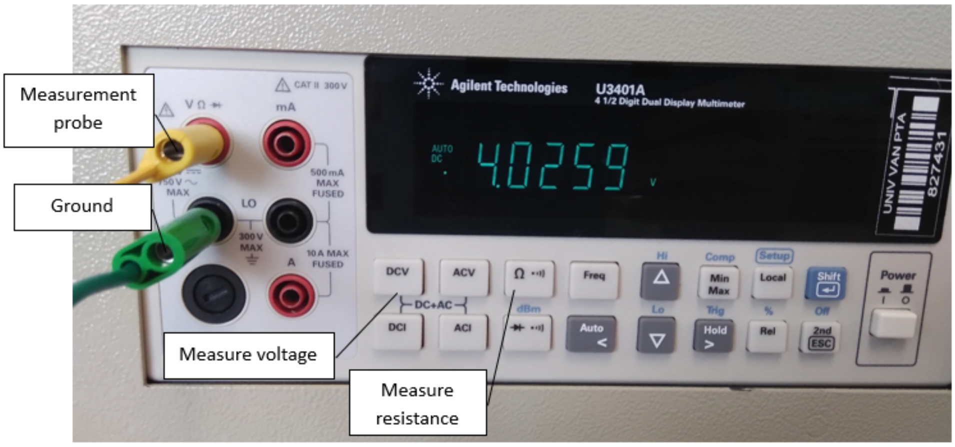

Figure 89: Connectors attached to the multimeter to measure voltage.

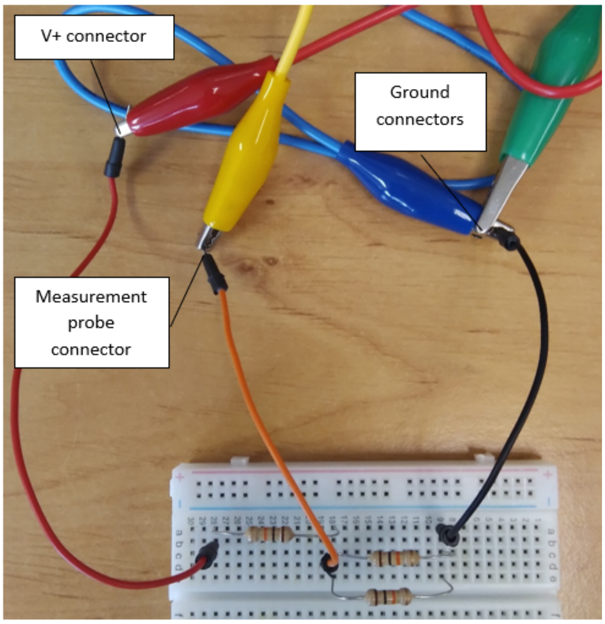

Figure 90: Connector wires from the power supply (Figure 88) and multimeter (Figure 89) attached to the circuit from Figure 87.

3.3.2 AC Circuits

If an AC voltage is supplied to the circuit, then the following steps are performed:

1. A BNC cable is connected to the output of the function generator and the first measurement port of the oscilloscope as seen in Figure 91. To activate the measurment channel, click the button with the '1' on it until it lights up.
2. The function generator is set up by pressing the 'wave gen' button which will take you to the screen showing the properties of the generated AC wave. The properties are adjusted by clicking on the corresponding buttons under the screen and turning the 'adjust all values' knob.
3. If the displayed waveform is too small or just looks like noise, click the 'Quick scale' button on the top right. When taking readings use the 'Run/Stop' button to see the wave in real time or pause it to measure points on the screen.
4. The BNC cables are connected as shown in Figure 92. The red side of the output of the wave

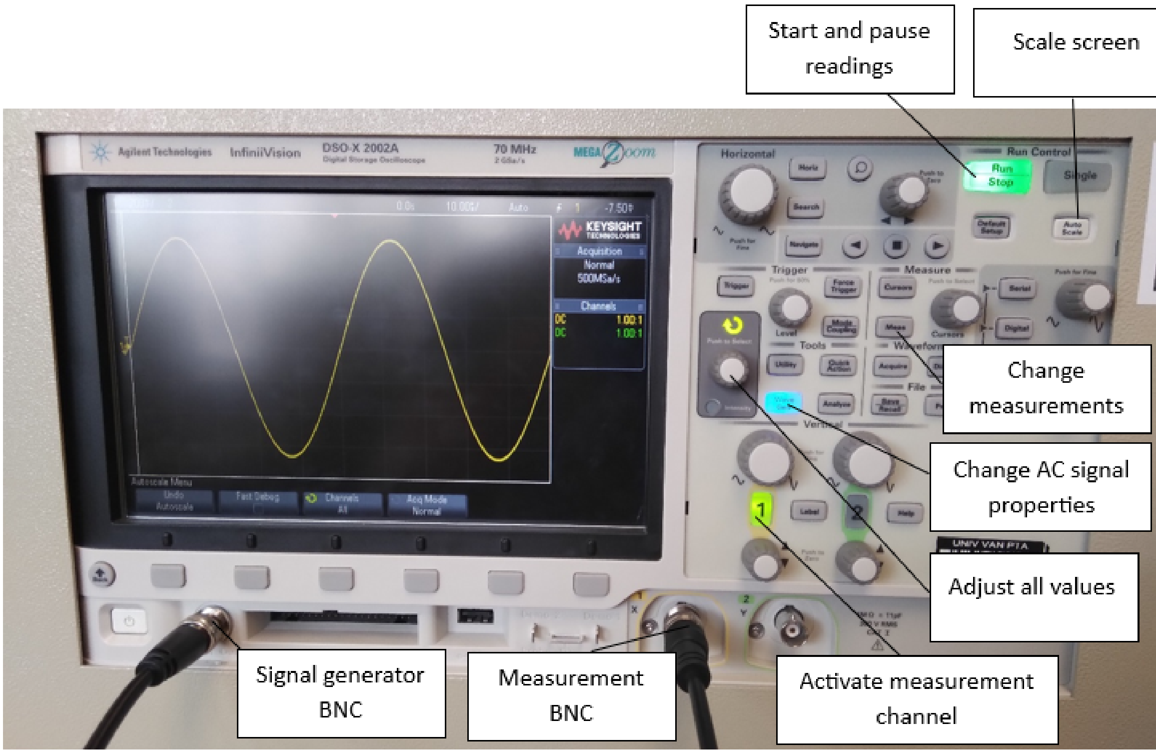

Figure 91: BNC cable connections and buttons used for an AC analysis.

generator is connected to the red wire. The red side of the channel 1 BNC is connected to the orange wire. Both black cable are connected to the ground node. It is also good practice to connect both to the COM port of the power supply to reduce noise in the measurements.

5. The oscilloscope is set up by pressing the 'meas' button as shown in Figure 91. The screen will then show what can be seen in Figure 93.

6. Start by clearing all measurements using the 4th button from the right. Then select the measurement type that is requested for the practical. To move up and down the list, use the 'adjust all values' knob. To add the reading to the box at the bottom right of the screen, push the 'Add measurement' button. Repeat until all desired measurements are added.

7. To read a value on the graph, a probe can added by pressing the 'cursors' button. This will bring up the dotted vertical and horizontal lines on the screen. Use the knob next to the 'meas' button to change the location along the x-axis. The horizontal line will move with the curve of the wave and its value can be read off the top of the screen.

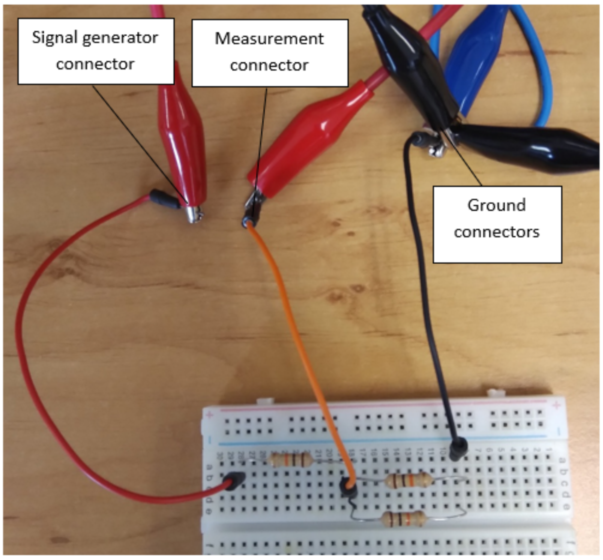

Figure 92: BNC cables from the oscilloscope and function generator connected to the circuit in Figure 87.

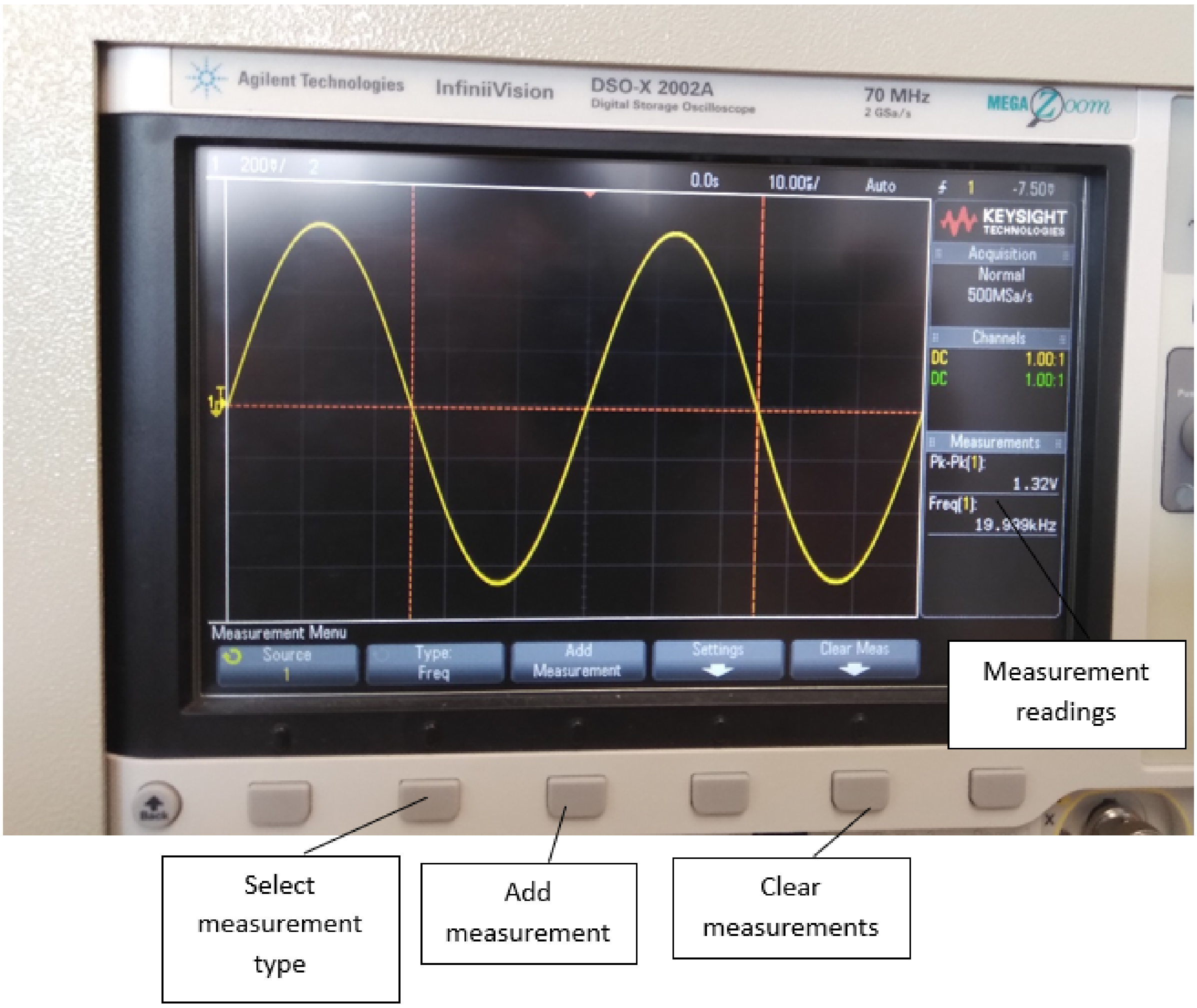

Figure 93: Layout of the measurement functions of the oscilloscope.